

RESEARCH PROJECT 1

Modeling Spatial Coordination and Heterogeneous Experience in an Emergency Department

Technical report

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Abstract

Space defines how people see, where they can move, and whom they can encounter. As sensors, digital models, and simulation make these relationships increasingly measurable, physical spaces can be studied as interfaces, with proposed alterations tested before construction. This study explores that possibility in an emergency department, where coordination depends on the interaction between spatial layout, clinical activity, and individual behaviour.

A spatial agent-based model of the University Hospital Zurich Emergency Department was developed using its floor plan, operational information, and 359 located interactions recorded during staff shadowing. The study proceeded through three scales. First, the baseline model was compared with the observed composition and spatial distribution of interactions. Second, it tested transparent partitions around the central coordination station, relocation of a smaller nurse station immediately south of it, and their combination. Third, a bounded generative-agent extension introduced five simulated personas with different priorities. A language model handled a fixed sample of their optional encounters, while recent contacts and accumulated experience informed later decisions, and subsequently appraised their completed shifts.

The baseline reproduced broad interaction hotspots better than exact locations and performed better than random spatial controls. Simulated focal interaction rates were far below the observed rates, and several details of who interacted and why differed. Within the model, transparent partitions added 9.07 interactions per hour under normal load, an increase of 39.7%, and 7.07 per hour under high load, an increase of 34.6%. Almost all of this increase came from additional contact between staff, while patient-facing contact remained nearly constant. Moving the smaller nurse station had little effect. Under the combined intervention, walking increased, and the coordination nurse accounted for 60% of the additional distance under normal load and 73% under high load. The five personas experienced similar gains in time with a visible colleague but initiated different amounts of additional contact. Under normal load, the change ranged from 0.03 to 2.22 contacts per ten agent-hours; under high load it ranged from 0.08 to 1.93. When 120 identical situations were replayed for every persona, their actions differed in 80% of situations and all five prespecified behavioural comparisons followed the expected direction. Their later appraisals also differed, with the same design supporting coordination, concentration, and overall person-space fit to different degrees.

In this study, the emergency department acts as a setting for spatial user research. Its central finding is not simply that visibility increased contact, but that a new sightline mattered only when existing work gave staff a reason to use it. Evaluating the design therefore required looking beyond total interaction counts to where contact occurred, how movement changed, who carried the resulting burden, and how users with different priorities responded to the resulting opportunities.

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1 Introduction

Coordination in an emergency department unfolds through motion. Rooms, corridors, and workstations open some routes and close others, shaping where staff meet and which conversations become possible. Yet more contact is not automatically better. A brief exchange may bring timely help, interrupt concentrated work, or expose a private conversation (Coiera et al., 2002; Kosits and Jones, 2011; Walter et al., 2017). The relevant question is therefore not simply whether a layout creates contact, but how a spatial opportunity becomes an interaction, where it occurs, and who bears its costs in movement and attention.

Human–Building Interaction (HBI) extends the concerns of human–computer interaction (HCI) from screens and devices to the environments through which everyday activity is organised (Alavi et al., 2019). Buildings have long been studied through post-occupancy evaluation, behavioural observation, and qualitative fieldwork. What is changing is the precision and scale at which their use can be recorded, modelled, and tested. Sensors can record aspects of how occupants use a building, digital models can connect those records to its spatial structure, and simulation can examine how that relationship might change under a different design. Together, these technologies allow rooms, buildings, and cities to be studied more like computational interfaces, with proposed alterations evaluated through models of user behaviour before the physical environment is changed.

An emergency department provides a demanding case for this approach because clinical work is distributed across people, tools, workstations, patient rooms, and established routines (Hollan et al., 2000; Hutchins, 2010). No single individual organises the department alone. Staff instead depend on what they can perceive, where they can move, and whom their current task makes relevant. A workstation may gather colleagues, a wall may keep them outside one another’s view, and a patient task may turn a visible colleague into a necessary partner. Spatial design is an arbiter of affordances, the bedrock through which activity is carried out.

Space syntax describes how this occurs by treating spatial configuration as a field of potential movement, co-presence, and encounter (Hillier, 1996; Turner et al., 2001). These possibilities are relational, as ecological psychology also asserts. An affordance is an opportunity for action available to a particular user with particular abilities and aims (Gibson, 1979). A sightline can therefore make a colleague visible without determining whether anyone will look, approach, or speak. That depends on who is present, what they are doing, and whether contact justifies interrupting the work at hand. In an emergency department, visibility may support awareness and help-seeking, but it may also invite distraction or expose private activity (Naccarella et al., 2019; Gharaveis et al., 2019; Ransolin et al., 2026). The study represents this process as a sequence: layout shapes movement and co-presence, visibility creates the possibility of co-awareness, ongoing work makes some colleagues relevant, and a contact decision determines whether that opportunity becomes an interaction.

A spatial agent-based model (ABM) connects this sequence by representing staff, space, work, and interaction within the same system. Each agent moves through the floor plan, performs role-specific tasks, encounters colleagues, and decides whether some optional contacts are worth pursuing. Repeating the same simulated shift under an altered layout then shows which differences the model attributes to the design change. The generative-agent extension adds another question: how might simulated users with different priorities respond to and appraise those differences? These comparisons remain bounded by the model’s cor-

respondence with the real department. Reproducing one observed pattern cannot establish that every mechanism or outcome is equally realistic (Windrum et al., 2007; Sargent, 2013).

Existing research has established several parts of this approach separately. Observational studies have connected emergency-department layouts with staff movement and communication, while empirically grounded multi-agent models have reproduced aspects of staff activity and dynamic visibility in healthcare settings (Tindle et al., 2020; Liu et al., 2024; Yuan and Zhou, 2025a,b). More recent work has begun using generative agents to examine navigation and occupant feedback in built environments (Noyman et al., 2026; Lee and Zhang, 2026). Taken together, this literature suggests that integrating movement, interaction, and heterogeneous user experience within a single emergency-department design workflow remains underdeveloped (Ransolin et al., 2026). This study contributes a proof of concept for such spatial user research. Located observations are used to evaluate the model's baseline, controlled interventions trace how a design change passes through visibility, movement, and interaction, and simulated personas examine how the resulting affordances may be experienced differently. Developing this proof of concept into a predictive design tool would require stronger behavioural validation and evidence connecting changes in staff coordination with patient care.

The study developed this approach across three progressively narrower scales:

1. How closely does the baseline reproduce the observed pattern of interaction across the Zurich emergency department?
2. Within the model, how do staff movement and interaction change when the central coordination station becomes visually open, when the smaller nurse station is moved northwards, or when both alterations are combined?
3. How do five model personas with different priorities respond to and experience these spatial changes?

The hospital floor plan identifies two principal workstations: **COCBIT**, the central coordination station, and **NURSTA**, a smaller nurse station immediately south of it. The study uses these names when describing the spatial interventions. Five model personas then introduce variation in how simulated staff balance coordination, interruption, patient focus, and vigilance when deciding whether to pursue optional contact. They represent contrasting patterns of behaviour within the model rather than measured personalities or categories of real staff.

Figure 1 summarises how the study moves from department-wide resemblance, through spatial intervention, to variation between simulated users. A language model contributes only to a fixed sample of optional-contact decisions and to the surveys and interviews generated after each shift.

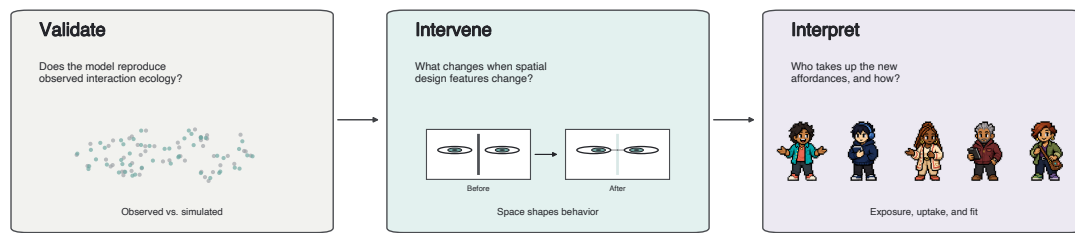


Figure 1. The study narrows from the whole department, to two changes in the floor plan, and finally to five different ways simulated staff may respond. Each part keeps the spatial and work rules established in the part before it.

Figure 2 introduces the modelled department, compares the observed and simulated spatial patterns, and identifies where the two interventions were made.

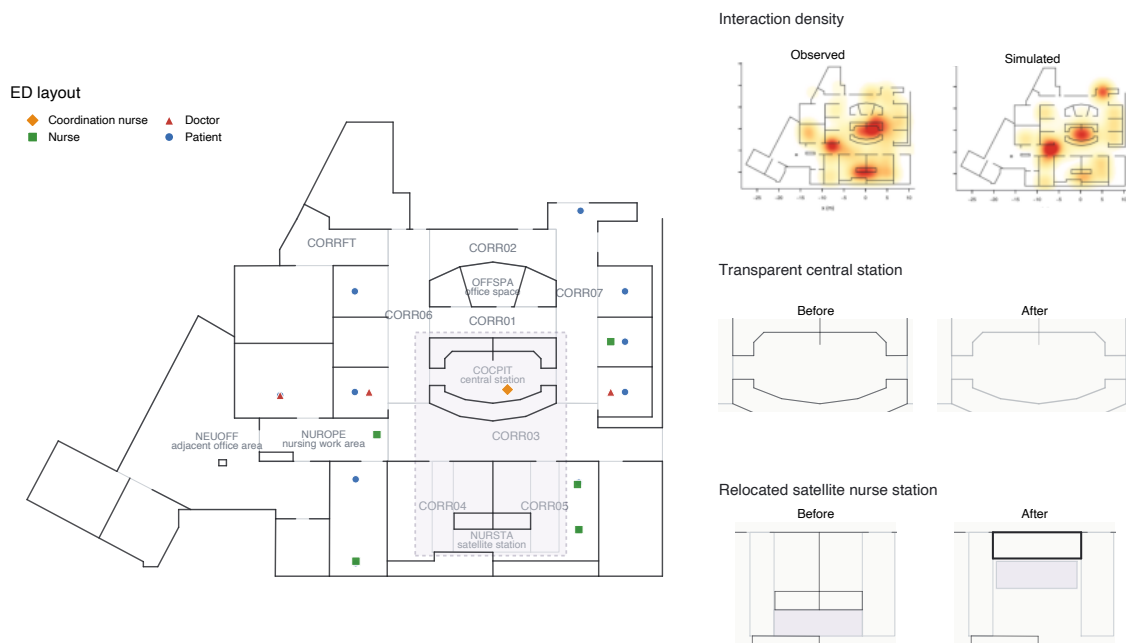


Figure 2. The modelled care area, observed and simulated interaction density, and the two design changes. The shaded dashed box marks the intervention area. The central-station change alters sight but not movement, while the satellite nurse station and its staff area are relocated within the southern clinical core.

Transparent partitions increased staff contact in the model because routine work brought relevant colleagues into the new sightlines. The gain came with additional walking, borne disproportionately by the coordination nurse. The five personas also responded differently to broadly similar gains in visibility. Together, these findings show why assessing a spatial change requires following it beyond geometry to the work and users that give it consequence.

2 Methods

2.1 Study design

A spatial agent-based model represented one emergency-department care area, including its walls, workstations, staff roles, and clinical tasks. The study comprised three linked analyses: Part 1 compared the baseline model with observed face-to-face interactions, part 2 held staffing, workflow, and behavioural rules constant while changing two parts of the floor plan, and part 3 repeated the spatial experiment with five model personas to examine variation between simulated users.

An interaction could occur only after a sequence of conditions was met. Two staff members had to be mutually visible, have a work-related reason to communicate, be available to act, and move within 1.5 m of one another. The contact rule then determined whether an optional opportunity was pursued. Only completed encounters were recorded as interactions (Figure 3).

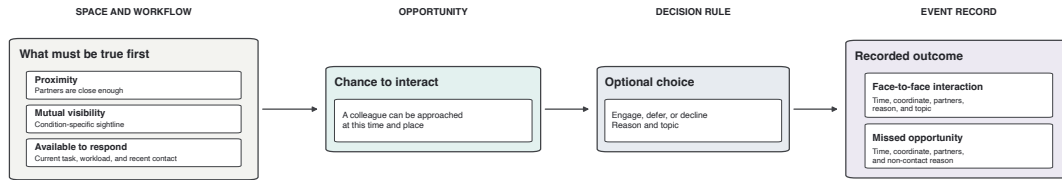


Figure 3. How the model produces an interaction. Staff must be able to see one another, have a work-related reason to speak, be free to act, and come within 1.5 m.

2.2 Observed interaction data

The empirical reference comprised event-level records collected during staff shadowing at the University Hospital Zurich Emergency Department in December 2023. The data were collected as part of the EVIDENT project and the study described by [Gath-Morad et al. \(2024\)](#). Participants provided informed consent, and the study received ETH Zurich ethics approval (EK 2023-N-31).

The file contained 610 records from six observation dates and 14 participants. Each record described an event involving the person being shadowed. The analysis retained 359 face-to-face interactions that identified an interaction partner and occurred within the modelled care area. The main resemblance analysis compared the composition and spatial distribution of these interactions with those produced by the baseline model.

A secondary analysis compared interaction rates for the shadowed person. Because formal observation boundaries were unavailable, the interval between each participant's first and last recorded events on a given date was used as an approximate observation window. Of the 359 retained interactions, 354 belonged to 14 participant-date fragments whose staff roles were represented in the model. These fragments covered approximately 12.19 person-hours. Simulated staff in the corresponding roles were observed for the same durations. Several empirical fragments were short, so the comparison was repeated using the four fragments

spanning at least one hour. The resulting rates were therefore treated as a denominator-sensitive diagnostic rather than calibrated estimates of interaction intensity.

Observed staff were mapped to three modelled roles: coordination nurse, nurse, and doctor. Interactions were then classified by partner role, topic, and broad location using fixed categories, while missing or ambiguous information remained unclassified.

The model was developed from the Zurich floor plan, internal information about the department, and the staged patient workflow used in EVIDENT's full-scale simulation studies. Early versions followed the restricted capacity and staffing of those studies but produced very few staff interactions. The number of beds and staff was therefore increased during development to create a busier care area and better reproduce the observed interaction patterns. Behavioural rules were refined through the same iterative process. These changes constituted manual model development rather than formal parameter calibration. Part 1 consequently measures the completed model's correspondence with observations that also influenced its development. The date-split analysis shows whether that correspondence persists across different observation days, but it is not validation against an independent dataset.

In addition to the missing observation boundaries, the available material lacked estimates of coordinate precision and inter-observer agreement. These omissions limit the interpretation of absolute interaction rates and fine-scale spatial correspondence.

Shadowed staff also completed an end-of-shift questionnaire covering workload, concentration, communication, teamwork, helpfulness, access to information, and noise. These responses were used only in a later comparison with the simulated post-shift ratings. Full exclusion criteria appear in Appendix A. Appendix Table 26 maps the observed questions to the synthetic ratings.

2.3 Spatial and operational model

The two-dimensional environment contained patient rooms, corridors, staff stations, work areas, and separate barriers to movement and sight. This distinction allowed a partition to become transparent while continuing to block movement. Agents travelled between work locations along routes that respected the physical walls.

The final model contained one coordination nurse, five nurses, three doctors, and ten beds. One bed was reserved for patients in the two most urgent Emergency Severity Index categories, ESI 1–2, while patients classified as ESI 3–5 used the remaining nine. This allowed urgent arrivals to enter the care process and create immediate escalation and staff-priority demands. Patients progressed through placement, nursing assessment, medical evaluation, diagnostics, treatment, reassessment, disposition, and discharge or admission. Role permissions determined which staff members could perform each task. Task execution and communication were logged separately, and only explicitly logged communication counted as face-to-face interaction.

The model remained substantially narrower than the department it represented. Internal materials describe approximately 125 patients per day, a mean ED stay of 263 minutes, 19 treatment-bay places, and eight fast-track places ([University Hospital Zurich project team, 2025](#)). Staffing tables list four to six assistant doctors and two senior doctors in the clinical core during several shifts, in addition to staff working in other medical areas and the wider nursing and support system. By comparison, the model contained nine staff, ten beds, fewer roles, and a compressed set of workflows and microtasks. Its normal-load runs gener-

ated approximately 64 arrival attempts during each 12-hour simulation, although capacity constraints meant that not every attempt became a patient treated within the model.

The perception layer could register colleagues up to 18 m away within a 210-degree field of view when no wall blocked the sightline. Optional contact operated over a narrower range: only visible colleagues within 10 m could become candidates for interaction. Every ten seconds, the model assessed whether a candidate was relevant to the agent's current work. Possible reasons included handover, an update following a task, bed-flow coordination, or a brief station check-in. An approach could be prevented when either agent was busy, the pair had interacted recently, the area was crowded, or attention was limited. A selected encounter was recorded only if the agents moved within 1.5 m of one another. The resulting log recorded who met, when and where they met, the reason for contact, and its topic.

Each run represented 12 hours in one-second steps. Because every run began with an empty care area, the first two hours served as an initialization period during which patients and unfinished work accumulated. All reported behavioural outcomes, including interactions, opportunities, movement, visibility, and zone transitions, were calculated over the final ten hours. This window better represented an occupied department than the artificial beginning of the simulation.

The high-load/high-acuity scenario combined a stronger time-varying arrival stream with a greater proportion of urgent patients, increasing the share classified as ESI 1–3 from 55.0% to 64.8%, where lower ESI values indicate greater urgency. As demand rose, discretionary contact was suppressed earlier and more strongly, while ESI 1–2 cases could pre-empt eligible lower-priority staff activity. This condition was designed to test the model under sustained demand and urgent interruption rather than reproduce an empirically observed workload state. Its pressure index was an internal control variable and should not be interpreted as a validated clinical measure of workload.

The agents were bounded models of activity rather than general models of clinical thought. They represented only the role permissions, patient stages, routes, and communication reasons required for the experiment. Appendix B summarises the model's entities, workflow, principal parameters, and pressure rule using the ODD framework (Grimm et al., 2010). The executable specification in the repository provides the remaining implementation detail.

2.4 Part 1: baseline resemblance

Part 1 used 100 normal-load baseline runs with different random seeds. This replication count was selected by examining the cumulative mean and standard error of the simulated department-wide interaction rate as runs were added. Both had largely stabilised by 100 runs (Appendix Figure 14). Observed and simulated events were processed using the same definitions of role pair, topic, zone, duration, and spatial density. Separate implementation checks verified that patients progressed through the workflow, routes respected physical walls, recorded interactions satisfied the distance requirement, and all events contained valid coordinates.

Resemblance was first assessed by comparing the observed and simulated proportions of role pairs, topics, and broad locations. Spatial resemblance was then evaluated by smoothing each set of interaction coordinates into density maps and measuring the distance between them at bandwidths of 0.5, 1, 2, and 4 m. These scales ranged from relatively precise hotspot placement to broader spatial structure.

The six observation dates were divided into all ten possible pairs of three-day groups. For each partition, the two observed groups were compared to estimate how much the empirical spatial pattern varied across days. A simulated sample containing the same number of interactions was then compared with each observed group, and the two distances were averaged. This established whether the model approached the agreement found between different subsets of the observations, rather than requiring it to reproduce every recorded coordinate.

The same procedure was repeated with two random controls. Global random placement distributed interactions throughout the feasible care area, while zone-preserving random placement retained the observed number of interactions in each broad zone but randomised their positions within those zones. The second control tested whether the model recovered local hotspots beyond merely placing the correct number of interactions in broad areas. Formulas and complete diagnostics are provided in Appendices A and C.

2.5 Part 2: spatial intervention experiment

Part 2 tested two design changes that could each be absent or present. **COCBIT transparency** removed designated partition segments from the visibility layer while preserving physical collision and route geometry. **NURSTA relocation** moved the satellite station, its enclosing geometry, zone semantics, workstation attractor, and route waypoints. Crossing the two yes-or-no changes produced four conditions: Baseline, COCBIT only, NURSTA only, and Both.

Both changes came from the Zurich project context and were encoded from the available floor plan and operational information. The satellite station sits south of the central coordination station, near the lower edge of the clinical core. Its relocation moved the workstation, staff area, and associated routes.

Each condition was run under normal and high load for 100 seeds (800 runs). The same replication count was used in each cell. For a given scenario and seed, all four conditions received the same externally generated patient-arrival stream. Each intervention run could therefore be paired with a baseline that began with the same demand, reducing unrelated run-to-run noise (Kleijnen, 1988).

The geometric check combined direct measures, such as visible area, visible beds, route distance, and sight between stations, with visibility-graph measures from the space-syntax tradition: connectivity, visual integration, and visual intelligibility (Hillier, 1996; Turner et al., 2001). A separate local sensitivity analysis varied four active settings: the probability of responding to a contact opportunity, the crowding level that discouraged an approach, the delay before another contact with the same person, and the chance of returning to a station. Crossing parameter levels with scenario, condition, and 20 seeds produced 1,280 runs that passed all integrity checks. Full settings and secondary outcomes appear in Appendix D. The analysis maps robustness within a neighbourhood of the manually selected defaults (ten Broeke et al., 2016).

A separate perception-envelope audit examined whether the COCBIT result persisted under different definitions of who could see and approach whom. Maximum sight distance, field of view, and approach radius were varied together across restricted, default, and expanded settings. Baseline and COCBIT only were compared under normal load using the same 20 seeds in each envelope (120 runs). The audit reported department-wide visibility, visibility among staff in the same named area, staff time with at least one visible colleague,

and the paired change in face-to-face interaction. Its bundled design supports one conclusion: whether the COCPIT effect persisted across three perception envelopes. The contributions of individual settings and uncertainty across the wider parameter space remain outside the audit's scope.

2.6 Part 3: language-model extension

2.6.1 Comparison with the observed questionnaire

Before the simulated end-of-shift ratings were interpreted, normal-load baseline responses were compared with the observed questionnaire. The questions covered maximum and critical workload, concentration, communication, teamwork, helpfulness to patients, information, and noise. Most used 1–7 scales. Critical workload used five percentage categories. The model has no acoustics, so noise served as an evidence-withheld control.

The language model used throughout Part 3 was Qwen3.6-35B-A3B, run locally on Snelius. For this preliminary comparison, it rated completed simulation summaries without seeing staff answers or clinical text. Normal-load synthetic responses were adjusted to the observed mix of roles and compared with 14 observed shifts for workload and 13 for the remaining supported questions. The comparison examined differences in group means, response distributions, and relationships among items, using the criteria detailed in Appendix G. For the 1–7 items, a mean difference within half a point counted as close. This was a descriptive plausibility check using the available observations. Independent validation would require new observations tested after the model and analysis were fixed.

2.6.2 Five model personas

Part 3 added five personas for optional contact: Team Connector, Focus Protector, Patient Advocate, Vigilant Monitor, and Adaptive Generalist. Their instructions gave different importance to coordination, interruption, patient focus, vigilance, adaptation, privacy, and initiative. They were designed to span contrasting responses to a familiar ED tension between coordination and interruption (Coiera et al., 2002; Walter et al., 2017). More broadly, they illustrate how a spatial model might compare the experience of selected user types in other buildings. The profiles were designed for this experiment rather than learned from staff. Five-factor OCEAN diagrams were later approximated from the written profiles to make those differences easier to recognise. They did not enter the prompts or analysis.

The experiment crossed two scenarios, four spatial conditions, ten seeds, and five assignment rounds (400 runs). Within each scenario and seed, every persona was assigned once to every staff position. The same simulated demand was also replayed across the four layouts. This matching lets differences be traced more cleanly to layout and persona rather than to a different mix of roles or patients. Ten seeds were the most that the language-model experiment could support, so the results report both their size and their run-to-run uncertainty.

The ABM continued to control movement, visibility, work, patient state, and which actions were possible. At a fixed 10% sample of optional staff encounters, Qwen chose only engage, defer, or decline, with a short reason and topic. Ordinary model rules handled the remaining 90%. The language model could change that sampled choice but could not invent an opportunity, alter the floor plan, or control movement. Each response was checked

automatically for the expected fields and linked to the unchanged situation that produced it.

For each choice, the language model could retrieve at most five completed interactions from the preceding two hours. Past events ranked higher when they shared a colleague, patient, work reason, situation, or location with the current opportunity. The persona also carried four bounded values describing coordination need, interruption strain, task continuity, and team support. Every two hours these values were updated from the persona's recent choices and completed contacts, then supplied to later decisions. Each update cited the events or choices that supported it. This allowed experience to accumulate across a shift without feeding generated prose back into the simulation. Missed encounters and post-run survey or interview answers were excluded. The design adapts memory retrieval and evolving agent state from generative-agent work while excluding autonomous planning, open conversation, and language-model control of movement (Park et al., 2023) (Figure 4).

The matched layouts held workload, demand, clinical role, and staff position constant, although a persona's earlier choices could still alter its later movement, timing, and available contacts. To isolate the persona policy itself, a separate construct check sampled 120 accepted decision situations from the completed runs and replayed each situation for all five personas. Within each five-persona set, the opportunity, task, recent contacts, and accumulated experience were identical, leaving the persona instructions as the only difference. Five directional comparisons were specified before inference to test whether each policy responded more strongly to the circumstances it was designed to prioritise. Paired bootstrap intervals were calculated by resampling situations. This design tests whether the implemented policies produce meaningfully different choices when given the same evidence.

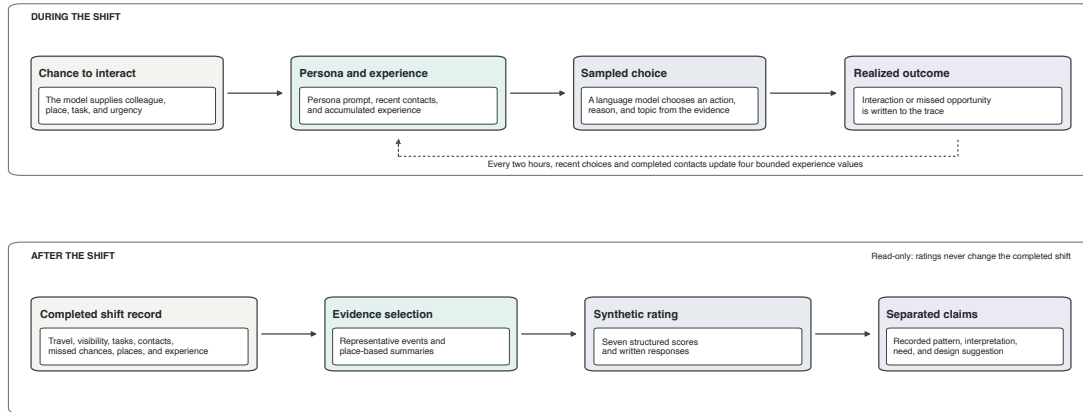


Figure 4. Where the language model enters the study. During a shift it chooses among three permitted responses for a fixed 10% sample of optional contacts, using the supplied situation, recent completed contacts, and accumulated experience. After the shift it describes the finished record without changing the simulation.

2.6.3 Post-run ratings and experience check

For each combination of workload, persona, and seed, one staff member was followed across all four layouts, producing 100 paired units and 400 staff-shift ratings. Each completed shift generated one structured survey and one six-question interview packet (800 verified responses). Packets compressed source-linked events from early, middle, and late shift phases,

place exposure, movement, mutual visibility, opportunities, realised and missed contact, workflow context, retrieved contacts, and final accumulated experience values.

The survey rated team awareness, interruption burden, task continuity, patient accessibility, privacy/control, spatial legibility, and overall person–space fit on anchored 1–7 scales, with null required when evidence was insufficient. Interviews asked about the overall shift, critical incidents, coordination and focus, supportive and difficult spatial features, and one suggested design change. Each answer had to identify which claims came directly from the logged shift and which were the persona’s interpretation or a proposed design idea. The events remained the evidence for the interpretation. This distinction only prevented an interpretation or suggestion from being presented as though it had occurred in the simulation. All answers were labelled synthetic.

A separate implementation check replayed 500 completed-run choices using the same persona, opportunity, and recent contacts after removing accumulated experience. It recorded whether the choice changed, thereby testing whether accumulated experience affected later decisions as intended.

2.7 Outcomes and statistical analysis

Primary Part 1 outcomes were role-pair, topic, zone, and spatial-density resemblance. Primary Part 2 outcomes were actionable opportunities, face-to-face interactions per hour, staff–staff and patient-facing shares, corridor and bedside shares, total movement, movement per staff contact, mean interaction separation, role–location segregation, mutual staff visibility, and role-specific movement and contact participation. Primary Part 3 outcomes were additional persona-initiated contacts and synthetic overall fit. The fit rating adapts the relational idea of person–environment fit (Edwards et al., 1998) to the completed simulated shift. To ask whether a layout served the personas broadly rather than only on average, inclusive fit reported three summaries: the average rating across the five personas, the lowest persona average (the “fit floor”), and the spread between personas. Privacy and control were not treated as a main outcome because only 38 of 400 rating packets contained enough evidence. Formal definitions and denominator rules are provided in Appendix C.

Part 1 reported average discrepancies in role pairs, topics, and zones, plus multiscale spatial distances tested across different splits of the observation dates. Part 2 compared each intervention run with the baseline run that shared its seed and arrival stream, then reported the mean difference and its 95% Monte Carlo confidence interval to quantify run-to-run uncertainty across paired simulations. Because COCPIT and NURSTA could each be either unchanged or altered, the analysis also asked whether doing both produced more or less than the two separate effects added together. Part 3 kept workload, simulated demand, persona, clinical role, and individual staff position matched across layouts. Differences were first averaged within each of the ten seeds. The results emphasise effect size, uncertainty, and whether the seeds generally pointed in the same direction rather than a large set of yes-or-no significance tests.

Because the intervention changed which optional encounters occurred, a secondary Part 3 diagnostic standardised Baseline and Both to the same mix of receiving staff role, work reason, and partner role within each workload and persona. Only strata present in both conditions were retained, and paired seed bootstrapping quantified run-to-run uncertainty. This controls coarse opportunity composition, not the complete situation. Engagement rates

are therefore reported in the appendix, while the common-situation construct check provides the stricter comparison of persona policies.

Qwen supplied both the sampled in-shift choices and the post-run ratings. The ratings could not change a completed simulation, but using the same model for both steps may make its choices and later explanations agree more neatly than independent human judgments would.

2.8 Synthetic interview audit and reproducibility

The design contained 40 combinations of workload, layout, and persona (2 scenarios $\times$ 4 conditions $\times$ 5 personas). One six-question interview was selected from each combination, giving 40 interview bundles and 240 answers. OpenAI’s Codex performed a separate audit after the runs that checked whether each statement was supported by its source, kept observation apart from interpretation, matched scores with prose, answered the question directly, and acknowledged trade-offs. No human rater independently reviewed or coded the answers. The audit retained 140 answers and excluded 100 that went beyond their source record. Repeated design suggestions were counted only in the 37 retained answers to the design-change question, so they remain exploratory synthetic hypotheses.

The project repository preserves the code, experiment plans, model revision, selected synthetic excerpts, aggregate outputs, automated checks, and figure and table builders.¹ Runs were accepted only when their workflow, routes, physical contacts, files, and expected counts passed the planned checks. Detailed integrity and rating audits are reported in Appendices E and F.

3 Results

3.1 Part 1: Baseline resemblance

All runs passed the checks for workflow, routing, interaction proximity, and coordinate integrity. The baseline was then compared with the observed distribution of interaction partners, topics, locations, and spatial density. These comparisons tested whether the model reproduced enough of the observed organisation of contact to support the subsequent layout experiment.

The model reproduced some categorical patterns but not all of them. It overrepresented doctor–nurse (+5.8 percentage points), doctor–patient (+4.7), next-steps and planning (+12.6), and bedside events (+7.0). It underrepresented coordination nurse–nurse (-3.9), patient-status updates (-8.1), informal or relational communication (-5.5), and corridor events (-2.9). Overall, role-pair and zone distributions were closer to the observations than topic distributions. The full comparison is reported in Appendix Table 10. The comparison is visualised in Figure 5.

The model did not reproduce the observed interaction intensity. Across all 14 participant-date fragments with a focal role represented in the model, the empirical extract contained 354 face-to-face records over 12.19 approximate person-hours, or 29.04 per hour. Applying the same roles and fragment lengths to simulated counterparts produced 5.92 per hour; across seeds, 95% of simulated rates lay between 4.58 and 7.25. Short empirical fragments

¹<https://github.com/arisaanto/ed-abm>

make the first-to-last-event denominator unstable, but restricting the calculation to the four fragments spanning at least one hour still gave 22.37 observed and 5.20 simulated interactions per hour.

The difference also reflects how events were represented. The empirical sheet divided shadowing into successive activity records. In 57 records, face-to-face contact was noted alongside computer, telephone, or device use; device-only activity was not counted as face-to-face interaction. More than half of the face-to-face records lasted under 30 seconds. A continuing episode could therefore generate several adjacent records as the partner, interface, task, or location changed. The simulation instead logged pairwise events only after an explicit work reason, availability, approach, and proximity gate, and it suppressed some rapid repeat contacts. It also omitted paramedics, transport, security, support roles, many clinical microtasks, and the technologies that structure coordination in the larger real department. The rate gap cannot be assigned to any one of these differences, so the remaining comparisons test structural resemblance rather than calibrated interaction frequency.

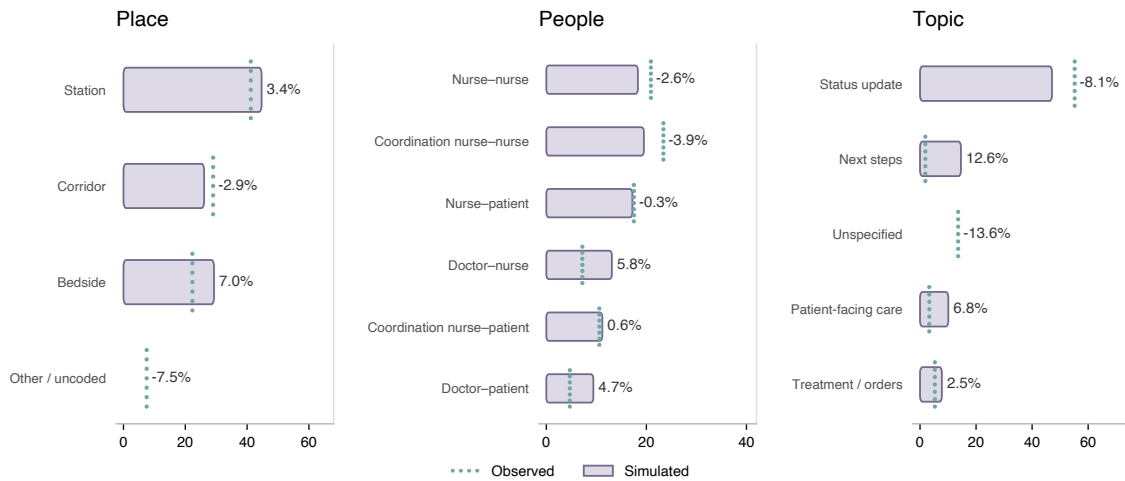


Figure 5. Observed and simulated interaction fingerprints under the normal-load baseline. The visual comparison exposes role-pair, topic, and zone resemblance and mismatch.

The spatial comparison asked whether interactions gathered in the same parts of the department. Observed and simulated locations were smoothed into density surfaces so that nearby points contributed to the same local pattern. The maps show the same main central and peripheral concentrations without requiring a one-to-one match between individual coordinates. Figure 6 shows the resulting point distributions and density surfaces.

Resemblance improved as the comparison moved from exact locations to broader spatial patterns. At a 0.5 m bandwidth, simulated samples were consistently farther from the observations than two halves of the observed data were from each other. At 1 m, the two distances were nearly equal. At 2 and 4 m, the model met or bettered this empirical repeatability benchmark in 9 and all 10 date partitions. Median model-to-observation distance fell from 0.554 at 0.5 m to 0.191 at 4 m and was lower than both random-placement controls at every scale. The baseline therefore reproduced broad hotspot structure better than fine-grained placement. Full comparisons are reported in Appendix Table 9.

Part 1 supports using the model to examine the direction and mechanism of spatial changes within the simulation. The mismatch in absolute interaction rate and the remaining categorical errors mean that its effect sizes should not be treated as calibrated predictions for the Zurich ED.

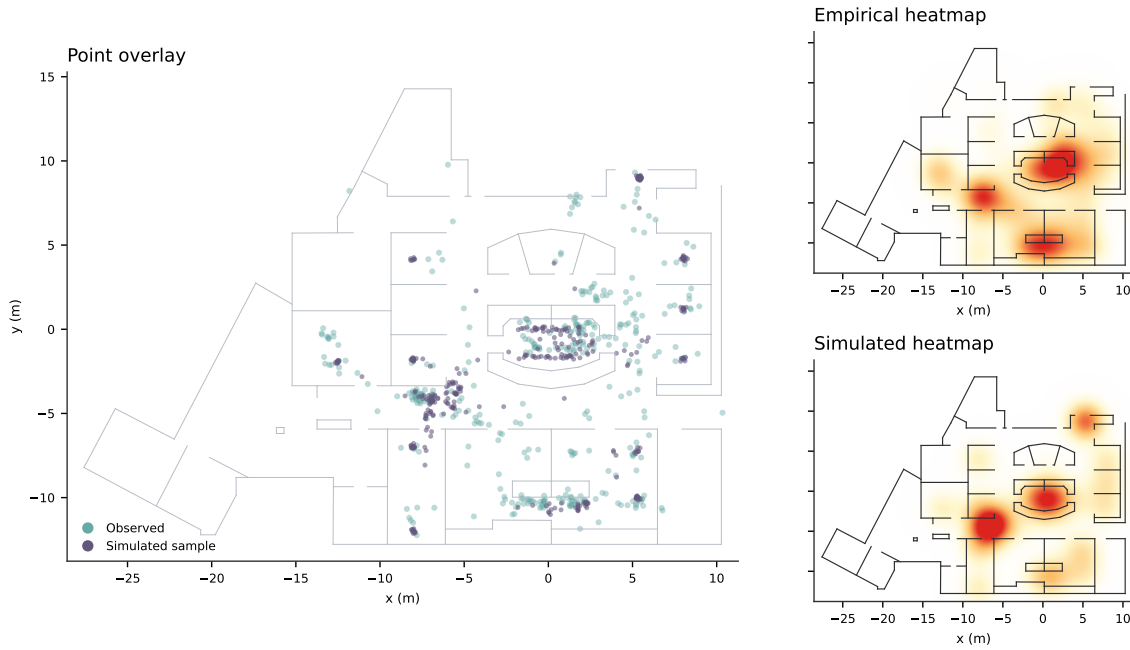


Figure 6. Observed and simulated interaction locations and smoothed density maps. The same broad hotspots appear, although their strength and exact location differ.

3.2 Part 2: Spatial interventions

3.2.1 Transparent partitions changed visual access and contact

The first change altered what staff could see. Making the central coordination station (COCPIT) partitions transparent expanded the floor area visible from the station from 37.78 to 224.10 m² and brought seven beds into direct view. The walking route to those beds stayed 12.83 m because the partition still blocked movement. Visibility also increased across the plan as a whole. Moving the smaller satellite nurse station (NURSTA), immediately south of the central station, left the same measures near baseline (Table 1 and Figure 7).

Table 1. Plan-wide visibility measures by spatial condition.

Metric	Baseline	COCPIT only	NURSTA only	Both
Mean visible grid points	50.9	81.6	50.9	81.2
Mean visible floor area (m ²)	103.1	158.4	104.3	159.0
Whole-plan visual reach (0–1)	0.404	0.534	0.404	0.532
Local view predicts whole-plan access (r)	0.763	0.953	0.766	0.953

Table 1 describes how each layout changed visual access across the floor plan. Mutual staff visibility was measured separately as the percentage of all staff-pair seconds during the final ten-hour analysis window in which both staff members could see one another. Under normal load, COCPIT transparency increased this exposure from 1.04% to 2.55% (+1.51 percentage points). Under high load, it increased from 0.72% to 1.80% (+1.08 points). These percentages use all staff-pair seconds as the denominator, so they measure department-wide exposure. Appendix Table 15 reports all conditions.

With the new sightlines, the model generated more work-relevant opportunities and completed contacts. COCPIT transparency created 545.7 additional chances to approach a colleague per normal-load run and 435.1 under high load. Realised face-to-face interaction increased by 9.07 and 7.07 events per hour, respectively. Across seeds, those COCPIT ef-

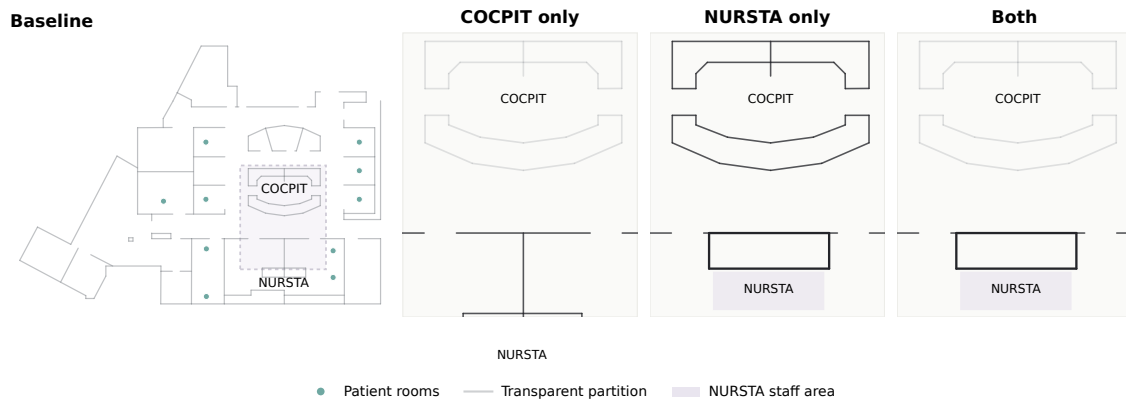


Figure 7. The complete intervention atlas. Baseline shows the full modelled floor plan and the shaded intervention area. The remaining panels show the central-station transparency change, the satellite-station relocation, and their combination.

fects had standard deviations of 2.27 and 1.83 events per hour. The combined condition produced changes of similar magnitude, whereas NURSTA relocation remained near zero. The difference between the combined effect and the sum of the two separate effects was -0.37 events per hour under normal load (95% CI $[-1.00, 0.26]$) and $+0.29$ under high load (95% CI $[-0.23, 0.81]$). Both intervals included zero, providing no clear evidence that NURSTA relocation modified the COCPIT effect. The combined effect was approximately additive. Exact paired contrasts are reported in Table 2 and Appendix Table 12.

Quantity, however, was only the first change. The new contacts also occurred differently. In the combined condition, staff–staff share rose by 10.3 percentage points under normal load and 12.4 under high load, while corridor share rose by 10.1 and 7.8 points. The patient-facing share fell by 12.4 and 14.1 points. Absolute rates clarify this compositional shift. Under normal load, patient-facing contact was 8.34 events per hour at Baseline and 8.39 under Both. Under high load, it was 9.52 events per hour in both conditions. Staff–staff contact rose from 14.48 to 23.44 events per hour under normal load and from 10.89 to 18.24 under high load. The decline in patient-facing share resulted from added staff coordination, while patient-facing contact changed little in absolute terms. Their duration, quality, and clinical consequences remain unmeasured (Figure 8).

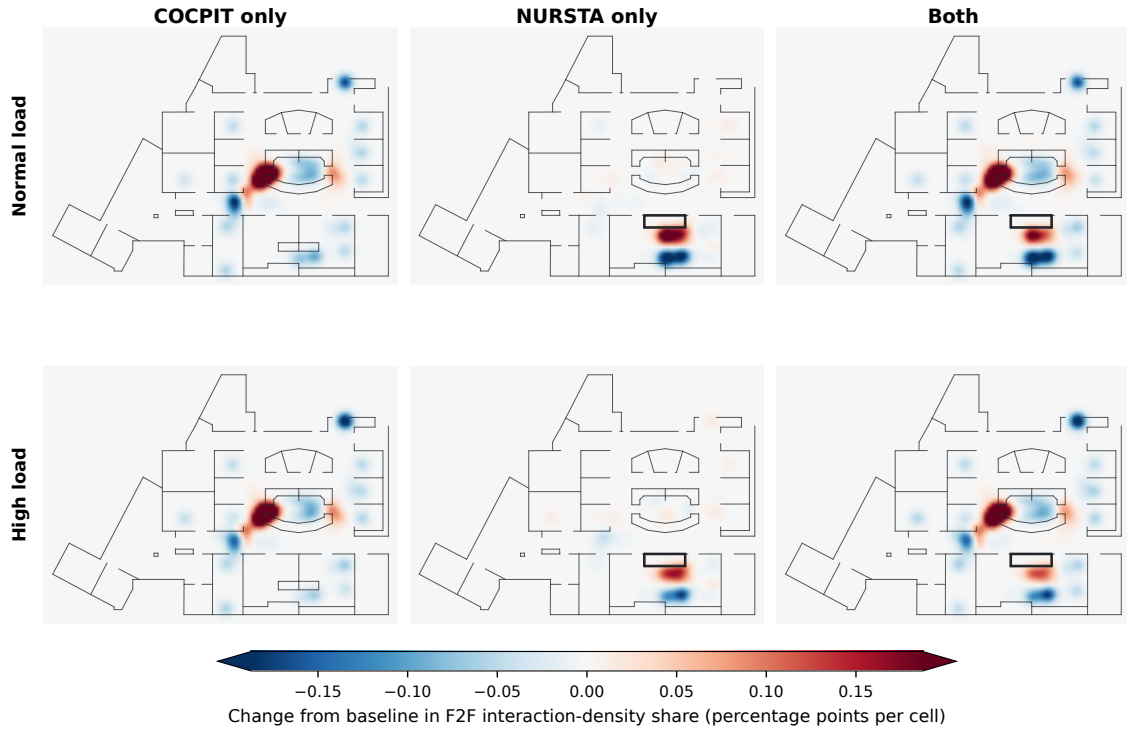


Figure 8. Redistribution of modelled face-to-face interaction relative to Baseline. COCPIT transparency increased staff contact and corridor use. The patient-facing share fell, but its absolute rate remained essentially stable.

Table 2. Principal paired intervention effects relative to Baseline.

Scenario	Contrast	Δ F2F/h [95% CI]	Δ staff-staff pp [95% CI]	Δ patient-facing pp [95% CI]	Δ corridor pp [95% CI]
Normal	COCPIT	+9.07 [8.62, 9.52]	+10.5 [9.8, 11.2]	-12.5 [-13.3, -11.8]	+10.5 [9.7, 11.3]
Normal	NURSTA	+0.31 [-0.08, 0.71]	+0.1 [-0.6, 0.8]	-0.4 [-1.2, 0.4]	+0.1 [-0.7, 0.8]
Normal	Both	+9.02 [8.55, 9.49]	+10.3 [9.5, 11.0]	-12.4 [-13.3, -11.5]	+10.1 [9.3, 11.0]
High	COCPIT	+7.07 [6.71, 7.43]	+11.9 [11.1, 12.7]	-13.7 [-14.6, -12.8]	+7.5 [6.6, 8.4]
High	NURSTA	-0.01 [-0.33, 0.30]	-0.3 [-1.2, 0.6]	+0.5 [-0.5, 1.4]	+0.5 [-0.3, 1.3]
High	Both	+7.35 [6.96, 7.73]	+12.4 [11.5, 13.2]	-14.1 [-15.0, -13.2]	+7.8 [6.9, 8.7]

Note. Paired seeds, $n = 100$ per scenario and contrast. Intervals describe Monte Carlo uncertainty across paired runs, not uncertainty about the real ED. Percentage-point effects are compositional. Bold estimates have 95% intervals excluding zero.

3.2.2 Coordination gains carried uneven walking costs

The coordination gain carried a measurable walking cost. During the final ten-hour analysis window, the combined condition increased staff movement by 1.1 km under normal load and 0.8 km under high load. Movement per staff contact nevertheless fell from 62.3 to 42.8

m under normal load and from 82.4 to 53.2 m under high load. Interactions also became more spatially concentrated and less separated by role, while staff spent more time able to see one another. The intervention combined greater total movement with more staff contact per metre travelled (Figure 9 and Appendix Tables 13, 14, and 15).

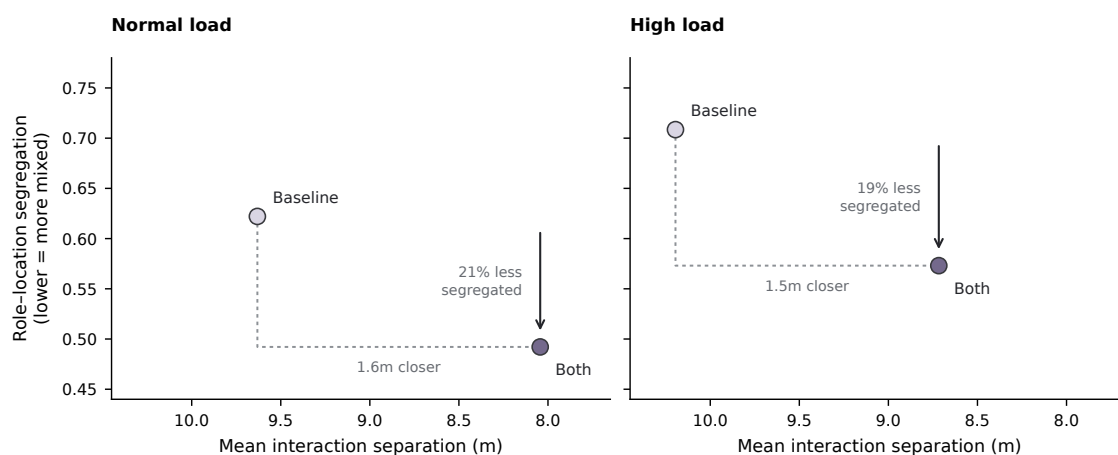


Figure 9. The combined intervention brought interactions closer together and made role pairs less separated by location. Points show condition means, and arrows connect Baseline with Both.

This gain was unevenly produced. The coordination nurse supplied 60.2% of the added movement under normal load and 72.8% under high load. The same intervention that reduced movement per staff contact thus intensified the modelled mobility and contact burden of the role already responsible for bed-flow coordination (Table 3).

Table 3. Role-specific decomposition of the combined intervention's added movement and staff contact.

Scenario	Role	Δ movement/person km [95% CI]	Share of added movement (%)	Δ contacts/person-hour [95% CI]
Normal	Coordination nurse	+0.64 [0.61, 0.67]	60.2	+6.31 [6.06, 6.56]
Normal	Doctor	+0.04 [0.03, 0.05]	10.8	+1.30 [1.18, 1.43]
Normal	Nurse	+0.06 [0.05, 0.07]	29.1	+1.54 [1.45, 1.64]
High	Coordination nurse	+0.56 [0.53, 0.60]	72.8	+5.50 [5.27, 5.72]
High	Doctor	+0.00 [-0.01, 0.01]	0.6	+0.91 [0.81, 1.02]
High	Nurse	+0.04 [0.03, 0.05]	26.6	+1.29 [1.20, 1.39]

Note. Both minus Baseline, paired by seed ($n = 100$ per scenario), with 95% confidence intervals in brackets. Movement share decomposes the total added movement. Bold identifies the role supplying most of that increase.

NURSTA's new location offered few opportunities for staff to use it during their modelled work. The sensitivity analysis tested whether this result depended on the chosen behavioural settings. Across 1,280 integrity-checked runs varying response probability, crowding threshold, repeat-contact delay, and station-return tendency, COCPIT transparency consistently increased contact while NURSTA relocation had little effect (Appendix Figure 15 and Table 16).

A separate visibility audit traced the mechanism from proximity to potential visual awareness to contact. It measured how long staff pairs were within approach range, how often those pairs could see one another, and how often they interacted. In this 20-seed audit, the COCPIT increase grew from 6.57 events per hour under restricted perception settings to 9.19 at the default settings and 11.90 under expanded settings (Appendix Table 17). Greater vi-

sual and approach reach thus amplified the modelled effect, while its direction remained consistent across all three settings.

3.3 Part 3: Simulated personas

3.3.1 Aggregate synthetic ratings resembled observed means

Normal-load baseline ratings were first compared with the observed end-of-shift questionnaire (Table 4). Across the six scored 1–7 items, the mean absolute error was 0.23 scale points. Five of six means fell within the prespecified half-point margin. Maximum workload differed by 0.54 points against the 0.50 margin. The critical-workload estimate was 25.5%, compared with 28.6% in the observed shifts. Four of seven item distributions met the compatibility threshold, one short of the prespecified requirement.

Table 4. Observed and synthetic end-of-shift questionnaire responses under normal load.

End-of-shift construct	Empirical mean (SD)	Synthetic mean (SD)	$ \Delta $
Maximum workload	4.57 (1.55)	4.03 (1.89)	0.54
Critically high workload (%)	28.57 (29.18)	25.54 (24.94)	3.04 pp [†]
Concentration	5.00 (1.15)	5.02 (1.43)	0.02
Communication satisfaction	4.77 (1.42)	5.07 (0.66)	0.30
Teamwork	5.15 (1.21)	5.04 (0.48)	0.11
Helpfulness to patients	5.15 (0.80)	5.28 (0.58)	0.12
Information sufficiency	5.31 (1.03)	5.01 (0.53)	0.30

Note. Empirical $n = 14$ for the two workload items and $n = 13$ for the remaining items. Synthetic estimates use 150 normal-load baseline staff shifts standardised to the observed role mix. Except where marked, scales run from 1 to 7. Bold differences meet the 0.5-point margin. [†]The Q2 mean is partly fixed by defining critical workload as the most demanding quarter of each simulated shift and is excluded from the six-item mean comparison.

Qwen rated the completed shifts without seeing the staff answers. Its ratings approximated the observed group averages, but synthetic answers for communication, teamwork, and information were more tightly clustered than the staff responses. Appendix G reports the full distributional and cross-item comparisons.

3.3.2 Five personas produced different responses to the same spatial change

Part 3 then asked whether the opportunities identified in Part 2 mattered equally to five model personas. Team Connector, Focus Protector, Patient Advocate, Vigilant Monitor, and Adaptive Generalist each occupied every staff position, allowing role, workload, demand, and spatial condition to remain fixed while the persona changed (Figure 10). Each persona is a simplified model personality that gives an agent different priorities when optional contact becomes possible. The OCEAN displays were approximated from those profiles to make the differences easier to recognise for the reader.

Interactive companion. <https://arisaanto.github.io/ed-abm/> presents scenario- and condition-specific quantitative profiles and audited synthetic excerpts for each persona. The interface presents model-generated evidence and does not contain human testimony.

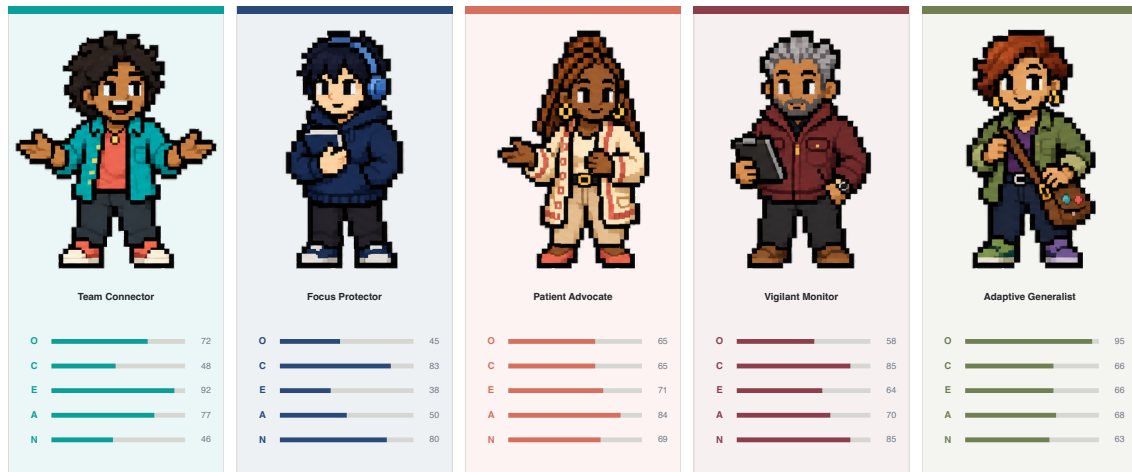


Figure 10. The five model personas used in Part 3. The OCEAN displays are plain-language illustrations approximated from their written profiles.

3.3.3 Similar visibility gains produced different amounts of additional contact

The combined intervention produced similar visibility gains across the five personas, ranging from 5.5 to 6.5 percentage points under normal load and from 4.2 to 4.7 under high load. Because earlier contact choices altered approach movement, pauses, and later positions, the gains were similar rather than identical. What the personas did with those conditions differed more clearly. Under normal load, the combined intervention added 0.03 persona-initiated contacts per ten agent-hours for the Focus Protector and 2.22 for the Vigilant Monitor. Under high load, the corresponding range was 0.08 to 1.93. The Team Connector, Patient Advocate, and Adaptive Generalist fell between those values (Figure 11).

The common-situation check separated policy differences from the opportunities generated by earlier choices. Across 120 identical situations, the five personas chose different actions in 96 cases (80%). All five prespecified comparisons followed the expected direction: the Team Connector engaged more than the Focus Protector in operational coordination, the Patient Advocate responded more strongly to patient-linked contexts, the Vigilant Monitor was more sensitive to urgency, and the Focus Protector and Adaptive Generalist became less willing to engage as spatial diversion increased. Full estimates appear in Appendix Table 22.

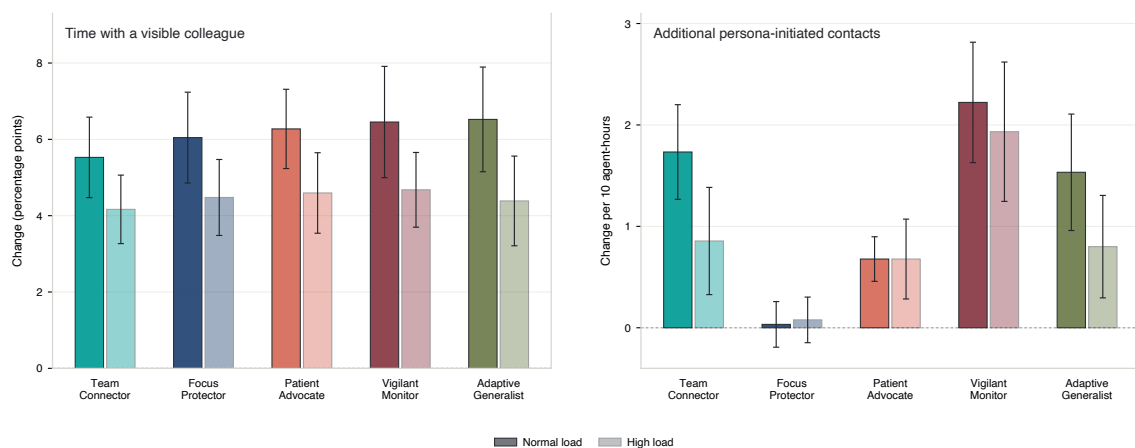


Figure 11. Realised visibility gains and additional persona-initiated contacts under the combined intervention. Bars show Both-minus-Baseline estimates with 95% confidence intervals. Contact estimates are reported per ten agent-hours.

A secondary engagement-rate analysis standardised receiving role, work reason, and partner role but did not make the complete situations identical. It is retained in Appendix Table 20 as a diagnostic of selectivity within common coarse opportunity types.

3.3.4 COCPIT transparency improved synthetic appraisals

Post-run ratings offered another view of the completed shifts. Under high load, the synthetic overall score increased from Baseline to Both for Team Connector (5.2 to 5.6), Focus Protector (6.1 to 6.3), Patient Advocate (4.3 to 4.8), Vigilant Monitor (4.6 to 4.9), and Adaptive Generalist (4.4 to 4.8). Table 5 shows these overall ratings alongside the dimensions from which they were drawn.

Table 5. Person–space fit across personas under high load. Scores show Baseline → Both on a 1–7 scale.

Persona	Overall fit	Coordination	Focus	Legibility
Team Connector	5.2 → 5.6	4.4 → 4.7	5.3 → 4.9	5.4 → 5.6
Focus Protector	6.1 → 6.3	4.2 → 5.2	6.3 → 6.6	5.1 → 6.2
Patient Advocate	4.3 → 4.8	3.5 → 3.9	3.8 → 4.9	4.8 → 5.6
Vigilant Monitor	4.6 → 4.9	3.6 → 4.3	4.2 → 3.8	4.9 → 5.7
Adaptive Generalist	4.4 → 4.8	3.8 → 4.3	3.8 → 3.6	5.0 → 5.9

Note. Scores average ten paired seeds, except the Adaptive Generalist focus score under high-load Both (nine rateable packets). Representative synthetic responses are reported in Appendix Table 25.

Across the five personas, the combined condition raised the average high-load rating by 0.4 points (95% CI [0.1, 0.7]). The lowest persona mean rose by 0.5 points, although its interval included zero [−0.3, 1.3]. Under normal load, the average gain was 0.3 points [0.0, 0.6], and the fit-floor change was also 0.3 [−0.6, 1.2]. COCPIT alone produced the clearest high-load appraisal change: average fit increased by 0.8 points [0.4, 1.2], the fit floor by 1.1 [0.2, 2.0], and the spread between personas decreased by 0.28 [−0.55, −0.02] (Figure 12 and Appendix Table 24).

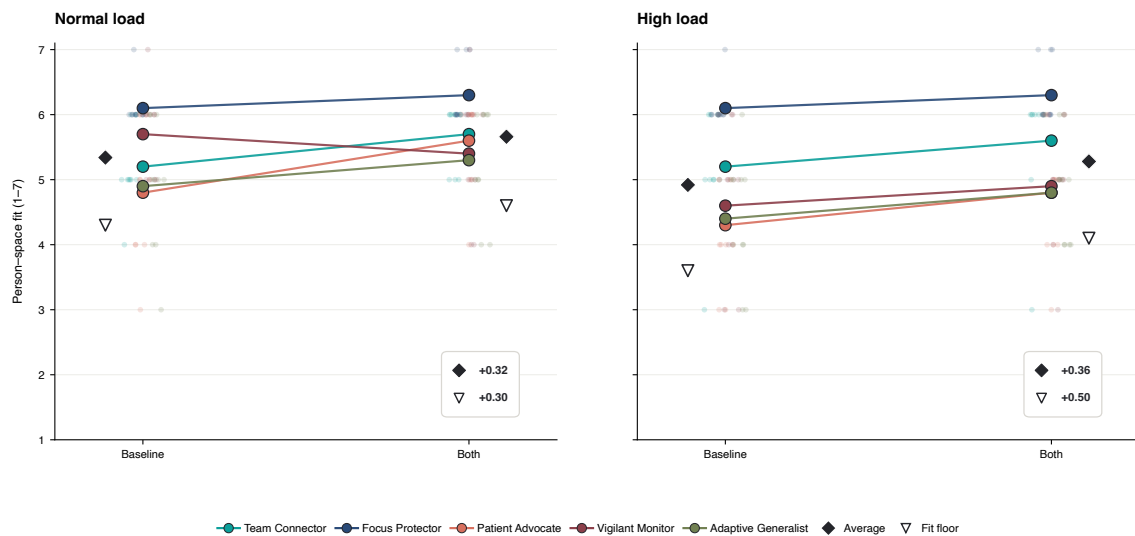


Figure 12. Person–space fit across personas. Trajectories connect Baseline with each intervention and report changes in average fit and the fit floor.

The fit floor is simply the lowest mean rating among the five equally weighted personas. It asks whether a higher average conceals one poorly served persona. It is not a measurement of accessibility or staff preference.

3.3.5 Accumulated experience changed some later choices

Removing accumulated experience from 500 otherwise identical choices changed 53 decisions (10.6%). Changes occurred for all five personas, ranging from 7% for the Vigilant Monitor to 16% for the Team Connector. Recent realized contacts and the current opportunity were held fixed. The check shows that the evolving experience values entered later decisions rather than merely appearing in the prompt (Appendix Table 21).

3.3.6 Synthetic interviews yielded design hypotheses

The audit reviewed 40 prespecified interview bundles containing 240 answers. It retained 140 answers and excluded 100 that went beyond their source records.

Among the 37 retained answers to the design-change question, 14 mentioned greater visual permeability, 14 a semi-private coordination space, 14 distributed support points, ten availability or focus signals, and three acoustic controls. Some answers raised several ideas together. They often paired better access to colleagues with ways to limit interruption and exposure: visibility was valued, while unrestricted openness also drew concern (Appendix Table 23).

Privacy/control could be rated in only 38 of 400 packets because the runs rarely contained direct privacy or acoustic evidence. Only three of the 30 paired contrast cells were complete, so privacy was not analysed as a main outcome. The retained answers offer synthetic design propositions for ED staff and hospital decision-makers to evaluate and refine.

4 Discussion

A useful user test should be capable of finding that an appealing design change does very little. Moving NURSTA changed the plan, but the modelled work seldom brought staff to its new location with a reason to coordinate. Opening the COCPIT partitions affected a station already woven into daily tasks. The contrast brings the challenge of testing a building into focus: a changed plan must be followed into the activities that give it consequence.

Space syntax has long described and tested how configuration shapes movement and encounter (Hillier, 1996). This study takes that work as a starting point and places ED tasks and interaction decisions alongside the spatial analysis. The model follows staff through a changed plan and records the steps between a new sightline and a completed conversation. It can also show when that sequence fails to develop, as it largely did after NURSTA moved. This is the model's value as a design probe. It makes assumptions about work explicit enough to inspect and gives a proposed alteration the chance to produce an unexpected or negligible result.

That probe has uneven credibility, which matters for how it should be used. The baseline recovered broad interaction hotspots better than fine locations and performed better than random spatial controls. Yet it produced far fewer face-to-face records than were observed and missed aspects of who spoke and why. A convincing-looking map can coexist with an incomplete account of daily communication. The current model is most useful for

investigating how its specified workflow converts spatial changes into contact. Its numerical results cannot yet forecast interaction rates in the Zurich department. A future spatial user-testing tool should make this distinction visible to its users: which behaviours have been checked against observation, which remain uncertain, and which design conclusions depend on them.

The increased contact around COCPIT raises a harder question than whether transparency “worked.” What did the additional conversations accomplish? Patient-facing contact remained roughly stable in logged frequency, while staff-to-staff contact grew. Care benefit remains unresolved. Observational research in an ED has found that coordination-related interruptions and exchanges containing less relevant information related differently to patients’ perceptions of care (Schneider et al., 2019). The content and consequences of an exchange matter. In this model, a conversation cannot yet improve a handover, change a clinical decision, prevent an error, or help a patient progress through care. The study reaches staff coordination and stops before the outcome that would ultimately justify a clinical redesign.

The location and cost of the new contact also deserve attention. More interaction occurred in circulation space, suggesting that a change centred on a workstation can reorganise communication beyond the workstation itself. Corridors carry clinical work alongside movement between rooms, as Carthey has argued for hospital design (Carthey, 2008). Whether these particular exchanges would be timely and useful, distracting, or too exposed for their content remains open. Their location gives future observation and staff review a more specific question than “Did communication increase?”

The department-wide gain was also carried unequally. Staff made more contacts per metre travelled, yet total walking rose, with the coordination nurse accounting for most of the added distance. An observational study of a UK ED likewise found that the nurse in charge carried a substantial communication load, although its setting and measures differ from this simulation (Woloshynowych et al., 2007). The model suggests a design question that an average interaction count would conceal: does easier access for the team create extra journeys and interruptions for the person holding its coordination together? Those journeys might be worthwhile. They might also make an already demanding role harder to perform. The results identify who would need to be observed and asked.

Part 3 extends this question from roles to users. The five personas encountered broadly similar gains in colleague visibility, yet they did not turn those opportunities into contact in the same way. The Focus Protector made almost no additional approaches while appraising some aspects of the changed space more favourably. The Vigilant Monitor approached more often while reporting a small loss of focus. Awareness itself may have value: a colleague can be easier to locate without every sighting becoming a conversation. Visibility may give one user useful information while drawing another into more demands. These are interpretations of designed model personas, but they show how much a single average response can conceal about a shared environment.

The common-situation check strengthens this part of the argument. When the personas received identical circumstances, their decisions still diverged in ways consistent with their stated priorities. Accumulated experience also affected some later choices. The experiment makes assumptions about different users operational, then tests their choices against the same situation. These five profiles are designed cases, with no claim to represent the Zurich workforce. The questionnaire comparison offers a narrower point of contact with staff data: synthetic group averages were close, while several response distributions were too com-

pressed. A mature version would need to learn from, and be challenged by, the breadth of actual staff behaviour and experience.

The synthetic interviews point to a design idea that the original transparent-versus-opaque intervention could not express. Their recurring theme was access to colleagues with some control over interruption and exposure. A semi-private coordination space, visual access, and availability cues combine those concerns in the concept shown in Figure 13. The image offers a concrete proposition for staff to contest. They may reject the problem it assumes, prefer another arrangement, or identify circumstances in which visibility should be adjustable. That exchange between a model-generated proposition and the people who use the building is itself part of spatial user research.



Figure 13. Speculative huddle-space concept generated from recurring synthetic interview suggestions. It is a prompt for staff co-design, not evidence of privacy preference, clinical benefit, or a recommended final design.

The larger ambition is to move some evaluation upstream, while changes are still proposals. This study demonstrates much of the path from layout to visibility, movement, contact, and heterogeneous simulated experience. The missing connection to patient care is substantial. A retail setting would need to connect spatial behaviour to shopping outcomes, a stadium might connect it to safe and comfortable movement, an ED must eventually connect coordination to care. Whatever the building, a meaningful test must follow behaviour through to the outcome the place is meant to support. At its strongest, this approach would ask whether a design helps different people accomplish that purpose, and who absorbs the costs along the way.

5 Limitations

The central limitation is the incomplete link between staff coordination and patient care. The model follows a design change through visibility, movement, and logged contact, while patient progress depends largely on preset tasks, service times, priorities, and role permissions. Conversations occupy staff time and may shift schedules. Their informational content cannot improve a handover, change a treatment decision, or prevent an error. Any resulting

change in simulated patient timing would arise from those scheduling rules without showing whether additional coordination helps care. The study identifies a possible reorganisation of work. Whether patients benefit remains unanswered.

The human evidence covers one ED, six observation dates, and a small number of shadowed staff. These observations informed manual model development and the later baseline comparison. Dividing the dates tests consistency within that material; the finished model has not been tested against an untouched shift or another department. The extract records located events without continuous staff trajectories or a measure of who could see whom. Movement burden and visibility changes consequently lack direct observed benchmarks. Formal shadowing-session boundaries were unavailable, making the observed interaction-rate denominator approximate. Some topic and location records were ambiguous, and the available material did not establish coordinate precision or inter-observer agreement. No ED worker evaluated either proposed layout.

Several assumptions shape how the interventions behave. Mutual visibility is a prerequisite for optional contact in the model, so transparent partitions tend to create more candidate encounters by construction. Tasks, availability, approach, and contact decisions determine how many become conversations. Staff can change their paths and timing within the model's rules, yet they cannot develop new routines around a relocated station. The model also compresses much of a real shift: telephone calls, documentation, monitoring systems, support staff, family conversations, informal exchanges, breaks, fatigue, and familiarity within the team. These omissions may contribute to the low interaction rate and are particularly relevant when interpreting NURSTA's small effect.

The high-load condition was designed to test heavier demand and urgency within the regular workflow. Major emergencies and breakdowns in that workflow were outside its scope. The two-hour warm-up reduced the influence of an empty starting department without establishing formal steady state. Behavioural settings were developed manually from the available information; they were not estimated from independent observations. Local sensitivity tests and three bundled perception settings showed that the main COCPIIT contrast persisted across the settings examined. They cannot bound the full uncertainty in staffing, workflow, geometry, and behaviour. Without observed staff-visibility data, none of the tested perception settings can be identified as the closest representation of the Zurich ED.

Part 3 used five designed personas, ten seeds, one language model and prompt design, and language-model decisions for 10% of optional contacts. The baseline questionnaire comparison gave some reassurance about group averages, although several synthetic response distributions were too narrow. It did not test how actual staff would respond to the altered layouts. Qwen supplied both sampled choices and post-shift appraisals, which may make those outputs unusually consistent with one another. The accumulated-experience values affected some later decisions but have no direct psychological validation. Privacy and acoustic evidence was sparse, and the synthetic interview audit had no independent human reviewer. Staff have yet to judge the generated design concept. These results demonstrate differences between the implemented persona policies; the range of experience among Zurich staff remains to be studied. The wider spatial user-research approach also remains untested beyond this department and its two proposed changes.

6 Future directions

The longer-term aim is a platform for spatial user research: a way to examine how different people use a place and test proposed changes before making them. The emergency department provides an initial case, but the approach should eventually work across settings with different layouts, users, activities, and goals. Reaching that point requires separating what can be reused from what must be learned about each new place.

The first step is to strengthen the Zurich model. Future observation should record complete shadowing windows and link the existing records of exchanges, their timing, location, and content, to the tasks underway and what happened next. Movement, occupancy, and lines of sight measured in the existing ED would help test the model's account of how staff encounter one another. A later shift or another department could then provide a genuinely held-out test. Most importantly, evidence connecting particular exchanges or interruptions to handovers, task timing, escalation, and patient flow would allow the model to examine whether a change in coordination affects care. Those relationships would need to be learned and tested, not assigned as automatic benefits of greater visibility.

A reusable system would have two connected parts. One would describe the setting: a two- or three-dimensional layout, barriers to movement and sight, technologies, roles, tasks, demand, and the outcomes that matter there. The other would describe its users, drawing on observations, interviews, and behavioural evidence instead of profiles invented solely for a simulation. This distinction would let the same modelling approach be adapted to an ED, shop, school, home, or transit hub while preserving the activities that make each setting different. Research on interview-grounded agents and models trained on human experimental responses suggests ways to improve the behavioural component (Park et al., 2024; Kolluri et al., 2025). Whether those methods can predict behaviour *in physical space* would still need to be tested with spatial observations and held-out users.

The product of such a system would be an interactive research tool, not only a set of summary figures. A researcher or designer could compare layouts, change an operating condition, follow particular users through a simulated day, and inspect when an opportunity became an action. They could ask how a design affected movement, contact, interruption, task completion, and user appraisal, then examine which groups gained or carried a new burden. The Zurich ED offers a concrete starting point for this interface: staff could view the four layout conditions over time, see how visibility and interaction changed, and interrogate the assumptions behind those changes. Synthetic suggestions, such as the huddle-space concept, could be brought to staff as prompts for co-design.

The same research loop could then be tested in other domains. In a shop, the questions might concern navigation, requests for assistance, and purchasing. In a home, they might concern when an embedded device is noticed, used, ignored, or experienced as intrusive. At the scale of a workplace, stadium, or city district, the relevant activities and outcomes would change again. Across these cases, the procedure would remain recognisable: observe people in context, build a model of the space and its users, test alternatives, and return the resulting predictions to people who can challenge them. The ambition is to make spatial and spatially embedded technologies testable as interfaces, while showing where a simulation has enough evidence to inform a design decision and where it does not.

7 Conclusions

This study brought spatial layout, clinical work, and simulated user decisions into one design test. The baseline captured broad interaction hotspots but produced far fewer face-to-face contacts than were observed. It gives the model a useful role as a probe of spatial mechanisms, while the size of its intervention effects remains to be tested with staff in the Zurich ED.

The two interventions show why changing a floor plan is only the beginning of that test. Transparent COCPIT partitions created sightlines that routine work repeatedly turned into staff contact. Moving NURSTA changed the layout without producing a comparable change in interaction. The added contact also came with more walking, much of it borne by the coordination nurse. Part 3 brought users into the same question: five designed personas encountered broadly similar gains in visibility, yet made different contact decisions and appraised their shifts differently. Their choices diverged even when they were given identical situations.

Together, these findings offer a proof of concept for spatial user research. A proposed design can be examined through what it makes possible, what people do with those possibilities, and how its benefits and burdens differ across users. This experiment stops before the question that matters most to an emergency department: whether changed coordination improves care. Connecting those steps with human evidence would make this approach more useful in Zurich and provide a foundation for testing spaces, and the technologies within them, across other settings.

8 Ethical Considerations

The Three Tier Model considers obligations to scientists, research participants, and society. Across all three, the central responsibility in this study is to keep observations of people distinct from simulated behaviour and generated accounts of experience.

8.1 Scientists

The report identifies which findings come from staff shadowing, which come from the agent-based model, and which are interpretations generated by Qwen. The baseline comparison used observations that also informed model development, so it is not presented as independent validation. The spatial interventions show changes in simulated coordination, while their effects on patient care remain unmeasured. Part 3 examines five designed personas, not established categories of clinicians.

Experiment manifests, paired seeds, model settings, aggregate outputs, and verification checks make the computational results inspectable. The synthetic interviews received a separate, rule-based Codex audit for grounding and overreach. No human rater independently coded them, and no inter-rater reliability assessment was performed. The audit is reported as model-assisted quality control, with answers that exceeded their evidence excluded from the reported qualitative findings.

8.2 Research participants

This project recruited no new participants and altered no clinical workflow. Its empirical reference is a coded extract from a consented University Hospital Zurich study approved by the ETH Zurich ethics committee (EK 2023-N-31). Even without names, detailed records of where and when people interacted may be indirectly identifying. Participant-linked records remain subject to the source study's access conditions and are not released in the public repository.

The synthetic personas raise a different concern. Their first-person answers can sound like testimony, although no clinician gave those answers. The personas were not derived from interviews with Zurich ED staff, and their OCEAN diagrams are illustrative profiles rather than psychometric measurements. Synthetic quotations are labelled accordingly. Any decision about an actual department would require the involvement of the people who work there.

8.3 Society

Spatial simulation could help teams examine costly design changes before construction and reveal burdens hidden by an average result. It could also give a narrow model more authority than it deserves. Optimising visible coordination alone might overlook privacy, concentration, patient access, and unequal demands on particular roles. Detailed movement and interaction traces could also be misused to monitor or rank staff.

This study offers decision support and testable design hypotheses. Its assumptions, uncertainties, and failures should remain visible wherever confidentiality permits, and its proposed changes should be assessed with affected workers and through post-occupancy evaluation. The model does not replace their participation or prescribe a design on its own.

9 Data and Code Availability

The project repository contains the model code, experiment manifests, selected verification materials, aggregate figures and tables, and the interactive persona explorer. Its README and results index explain what can be run or inspected without restricted data. The underlying shadowing records, questionnaire responses, participant-linked diagnostics, and complete simulation traces are not public. Access to the empirical material remains governed by the source study's ethics and data-management conditions.

10 GenAI Statement

OpenAI's Codex assisted with coding, debugging, analysis, verification, figures, repository organisation, literature discovery, and report drafting and editing. It audited 40 prespecified synthetic interview bundles for grounding. This was model-assisted quality control, not human review. OpenAI image generation supplied illustrative persona artwork outside the analysis. Qwen3.6-35B-A3B ran through vLLM on Snellius to make bounded Part 3 contact choices and post-shift ratings. These were synthetic outputs. The student remains responsible for all submitted claims.

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A Empirical data and baseline comparison

The extract held timestamps, coordinates, coded participant and sensor identifiers, staff roles, partner details, transmitted-information text, and location details for 610 events from six December 2023 dates. Fourteen participant identifiers formed 15 participant-date combinations. Of these records, 359 met Part 1’s face-to-face and partner criteria. Every record involved the shadowed participant. Formal session boundaries were unavailable, so absolute frequency served only as a denominator-sensitive diagnostic. The primary focal-rate calculation used all 14 role-compatible fragments (354 contacts over 12.19 approximate person-hours). A sensitivity calculation used the four fragments lasting at least one hour (208 contacts over 9.30 approximate person-hours).

Observed roles were mapped to coordination nurse, nurse, and doctor. Partner roles came from a frozen mapping, and role pairs had no direction. Fixed rules classified topics, with ambiguous records left low-information or unspecified. Zone polygons, aliases, and a fixed fuzzy fallback assigned broad bedside, station/desk, corridor, and other/uncoded locations. The complete role, topic, and zone comparisons are shown in Tables 6, 7, and 8.

Table 6. Observed and simulated role-pair distributions.

Role pair	Observed	Simulated	Difference
Coordination nurse + Nurse	23.4%	19.5%	-3.9 pp
Nurse + Nurse	20.9%	18.3%	-2.6 pp
Nurse + Patient	17.5%	17.2%	-0.3 pp
Doctor + Nurse	7.2%	13.1%	+5.8 pp
Coordination nurse + Patient	10.6%	11.2%	+0.6 pp
Doctor + Patient	4.7%	9.4%	+4.7 pp
Doctor + Doctor	7.2%	6.5%	-0.7 pp
Coordination nurse + Doctor	3.3%	4.8%	+1.4 pp
Coordination nurse + Support	2.5%	0.0%	-2.5 pp
Nurse + Support	0.6%	0.0%	-0.6 pp
Nurse + Unknown	1.1%	0.0%	-1.1 pp
Patient + Support	0.8%	0.0%	-0.8 pp

Table 7. Observed and simulated topic distributions. Topic labels are approximate semantic classes rather than conversational-content validation.

Topic	Observed	Simulated	Difference
Patient status update	55.2%	47.1%	-8.1 pp
Next steps/planning	1.9%	14.6%	+12.6 pp
Patient-facing care	3.3%	10.1%	+6.8 pp
Treatment/orders	5.3%	7.8%	+2.5 pp
Bed/capacity management	5.3%	6.5%	+1.2 pp
Diagnostic findings	3.6%	6.0%	+2.4 pp
Patient flow/transfer	0.8%	5.5%	+4.6 pp
Administrative/logistics	4.7%	1.8%	-3.0 pp
Informal/relational	6.1%	0.6%	-5.5 pp
Unspecified/low-information	13.6%	0.0%	-13.6 pp

Table 8. Observed and simulated broad-zone distributions.

Zone group	Observed	Simulated	Difference
Station / desk	41.2%	44.7%	+3.4 pp
Bedside / patient room	22.3%	29.3%	+7.0 pp
Corridor	29.0%	26.1%	-2.9 pp
Other / uncoded	7.5%	0.0%	-7.5 pp

Table 9. Cross-validated multiscale spatial resemblance and random-placement controls. Lower distance indicates greater similarity to the observed density surface.

Spatial scale (bandwidth)	Empirical-to-empirical	Simulation-to-empirical	Global random	Zone-preserving random	Simulation met benchmark
0.5 m	0.538	0.554	0.657	0.668	0/10
1.0 m	0.426	0.425	0.527	0.543	4/10
2.0 m	0.335	0.298	0.408	0.430	9/10
4.0 m	0.249	0.191	0.294	0.324	10/10

Note. Values are medians across the ten unique divisions of six observation dates into two groups of three. For each partition, equally sized simulated and randomized samples were compared with each observed half in turn and the two distances were averaged. Each simulation and control value is the median of 250 paired samples. Global random placement draws from feasible care-area locations. Zone-preserving random placement retains the empirical counts in station/desk, corridor, bedside/patient-room, and other/uncoded groups. The final column counts partitions in which simulation-to-empirical distance was no greater than empirical-to-empirical distance. Distance is the square root of Jensen–Shannon divergence using natural logarithms. Zero denotes identical maps and $\sqrt{\ln 2} \approx 0.833$ non-overlapping maps.

Table 10. Observed–simulated categorical divergence. Lower values indicate more similar category shares.

Comparison	Full JSD	Producible-only JSD	Observed mass excluded
Role pair	0.040	0.015	5.0%
Topic	0.165	0.101	13.6%
Zone	0.042	0.004	7.5%

Note. Jensen–Shannon divergence runs from zero for identical distributions to one for maximally separated distributions. Full JSD retains all categories. The restricted calculation excludes observed categories the model cannot produce. The final column reports the observed mass removed by that restriction.

B Agent-based model specification

The model contained nine staff agents, ten beds, patients, polygonal zones, physical and visual wall layers, and a waypoint routing graph. One bed was reserved for ESI 1–2 patients, while ESI 3–5 patients used the remaining nine. Staff moved at 1 m/s with 0.2 m wall clearance. Perception used a 210-degree field of view, 18 m range, line of sight, distance-sensitive salience, relevance, and bounded attention. Only visible colleagues within 10 m could become optional contact candidates. Optional staff opportunities were evaluated every ten seconds and required operational reason, interruptibility, and feasible action. Recorded face-to-face contact required separation of at most 1.5 m.

Patients progressed through placement, nursing assessment, medical evaluation, diagnostics, treatment, reassessment, disposition, and discharge or admission. The coordination

nurse performed placement. Nurses performed assessment, diagnostics, treatment, and discharge/admission. Doctors performed evaluation, diagnostics, treatment, reassessment, and disposition. Under normal load, the ESI 1–5 probabilities were 0.010, 0.120, 0.420, 0.300, and 0.150. Under high load they were 0.008, 0.140, 0.500, 0.230, and 0.122, increasing the ESI 1–3 share from 55.0% to 64.8%. The high-load arrival multiplier was 1.15, compared with 0.76 under normal load. Its pressure threshold fell from 0.62 to 0.50 and low-priority suppression rose from 0.35 to 0.55. ESI 1–2 cases could preempt eligible lower-priority staff activity. These changes increased sustained demand and urgent interruption inside the regular workflow.

The model’s pressure index was

$$P = \min(2, 0.34O + 0.25W + 0.15T + 0.16A + 0.10D)$$

where O is ordinary-bed occupancy relative to capacity, W is waiting count relative to three, T is oldest current wait in hours, A is active high-acuity load relative to two, and D is the number awaiting first doctor contact relative to three. Components other than O were capped at 1.5. This equation governed model behaviour and served only as an internal pressure control, without clinical workload-scale status.

The repository contains the executable specification, frozen configuration, floorplan geometry, workflow transitions, and validation gates. The principal modules are separated into reusable model code (src), executable study workflows (scripts), and frozen experiment plans (manifests).

C Measures and estimands

Categorical resemblance used Jensen–Shannon divergence,

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \| M) + \frac{1}{2} \text{KL}(Q \| M), \quad M = \frac{1}{2}(P + Q).$$

Categorical calculations used base-2 logarithms, giving a scale from zero to one. Spatial-density distance instead used the square root of Jensen–Shannon divergence with natural logarithms, giving a maximum of $\sqrt{\ln 2} \approx 0.833$. Bandwidths of 0.5, 1, 2, and 4 m tested whether the conclusion depended on treating a hotspot as a desk-sized, room-scale, or broader neighbourhood. Date-split cross-validation averaged the two possible directions for every three-day partition. The same sample sizes, date splits, bandwidths, and distance calculation were used for global-random and zone-preserving random-placement controls.

Visibility-graph measures were approximated on a 1.0 m regular grid over the largest connected walkable component, using the visibility walls defined for each condition. Connectivity was the mean number of directly visible grid points. Isovist area was the mean 360-degree ray-cast visible area. Visual integration was mean normalized closeness centrality on the visibility graph, and visual intelligibility was the Pearson correlation between local connectivity and global visual integration across grid points. The sample contained 250 grid points in Baseline and COCPIT only and 253 in NURSTA only and Both. These measures are study-specific approximations, not formal DepthmapX outputs.

Coordination cost was staff movement in metres during the final ten-hour analysis window divided by staff–staff contacts in the same window. It is a descriptive movement-per-contact ratio, not clinical efficiency. **Mean interaction separation** was the mean Euclidean

distance across all unordered pairs of interaction coordinates. **Role–location segregation** was normalized mutual information,

$$S = \frac{I(R; Z)}{\min\{H(R), H(Z)\}},$$

where R denotes role pair and Z broad interaction zone. Conditions were downsampled repeatedly to equal event counts within each scenario–seed quartet before estimation.

Mutual staff visibility was reciprocal-perception staff-pair seconds divided by all possible staff-pair seconds during the final ten-hour analysis window. It is an unconditional department-wide exposure measure. It does not condition on distance, shared location, or presence at a workstation and should not be read as the chance of seeing a colleague who is locally co-present. **Engagement rate** was sampled model-selected engage decisions divided by sampled eligible decisions within matched strata. **Average fit** was the mean of the five persona-specific mean scores, **fit floor** the smallest of those five means, and **persona disparity** their standard deviation. These are ensemble statistics over designed personas, not population accessibility measures.

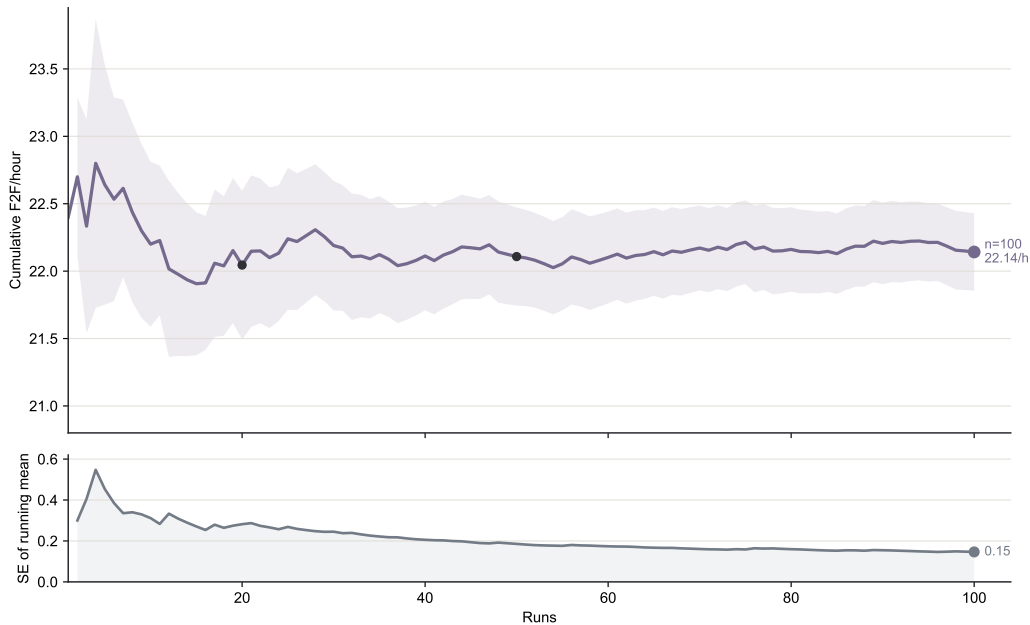


Figure 14. Replication convergence for the normal-load baseline. The upper panel shows the cumulative mean department-wide interaction rate across the final ten-hour analysis windows; the lower panel shows its standard error.

D Spatial intervention robustness

The following tables report the paired secondary contrasts and the local sensitivity design used to test whether the principal Part 2 result survived changes in active behavioural parameters. Movement and zone-transition contrasts appear in Table 11.

Table 11. Paired movement and zone-transition effects.

Scenario	Contrast	Δ movement km [95% CI]	Δ movement %	Δ zone transitions [95% CI]
Normal load	COCPIT only – Baseline	+1.3 [+1.2, +1.4]	+14.5	+336.1 [+313.7, +358.6]
Normal load	NURSTA only – Baseline	-0.3 [-0.3, -0.2]	-2.9	+76.3 [+55.1, +97.6]
Normal load	Both – Baseline	+1.1 [+1.0, +1.1]	+11.9	+428.1 [+406.0, +450.2]
High load	COCPIT only – Baseline	+1.0 [+0.9, +1.1]	+11.2	+271.1 [+250.7, +291.4]
High load	NURSTA only – Baseline	-0.2 [-0.3, -0.2]	-2.7	+80.7 [+62.6, +98.8]
High load	Both – Baseline	+0.8 [+0.7, +0.8]	+8.7	+359.2 [+340.2, +378.1]

Note. All outcomes use the final ten-hour analysis window. Contrasts use paired seeds ($n = 100$). Bold estimates have 95% confidence intervals excluding zero.

Table 12. Paired intervention effects on actionable interaction opportunities.

Scenario	Contrast	Δ opportunities [95% CI]
Normal load	COCPIT only – Baseline	+545.7 [510.4, 581.0]
Normal load	NURSTA only – Baseline	+11.6 [-18.8, 41.9]
Normal load	Both – Baseline	+544.3 [514.8, 573.8]
High load	COCPIT only – Baseline	+435.1 [416.3, 454.0]
High load	NURSTA only – Baseline	-1.0 [-14.1, 12.2]
High load	Both – Baseline	+443.7 [425.4, 461.9]

Note. Contrasts use paired seeds ($n = 100$). Bold estimates have 95% confidence intervals excluding zero.

Table 13. Absolute interaction rates and factorial interaction contrasts.

Scenario	Outcome	Baseline	COCPIT	NURSTA	Both	Factorial interaction [95% CI]
Normal	Total F2F/h	22.82	31.89	23.13	31.83	-0.37 [-1.00, 0.26]
	Staff-staff/h	14.48	23.56	14.69	23.44	-0.33 [-0.89, 0.23]
	Patient-facing/h	8.34	8.33	8.44	8.39	-0.04 [-0.31, 0.22]
High	Total F2F/h	20.41	27.48	20.40	27.76	0.29 [-0.23, 0.81]
	Staff-staff/h	10.89	17.93	10.82	18.24	0.38 [-0.07, 0.83]
	Patient-facing/h	9.52	9.55	9.58	9.52	-0.09 [-0.38, 0.19]

Note. Cell entries are condition means. The factorial interaction is Both minus COCPIT minus NURSTA plus Baseline, paired by seed. Brackets show 95% Monte Carlo confidence intervals across paired simulation runs, not uncertainty about the real ED.

Table 14. Coordination-efficiency profile by condition and scenario.

Scenario	Condition	Movement per staff contact (m)	Separation (m) [95% CI]	Segregation [95% CI]
Normal	Baseline	62.3	9.63 [9.57, 9.69]	0.622 [0.611, 0.633]
Normal	COCPIT	43.5	8.18 [8.11, 8.24]	0.495 [0.488, 0.502]
Normal	NURSTA	59.5	9.35 [9.30, 9.40]	0.611 [0.602, 0.620]
Normal	Both	42.8	8.04 [7.99, 8.10]	0.492 [0.486, 0.499]
High	Baseline	82.4	10.20 [10.14, 10.25]	0.708 [0.700, 0.717]
High	COCPIT	55.3	8.82 [8.76, 8.89]	0.586 [0.580, 0.593]
High	NURSTA	80.6	10.06 [10.01, 10.12]	0.702 [0.694, 0.710]
High	Both	53.2	8.72 [8.66, 8.77]	0.573 [0.567, 0.579]

Note. All outcomes use the final ten-hour analysis window. Bold identifies the lowest modelled coordination friction within each scenario, not a clinical optimum.

Table 15. Visibility, participation, and spatial-footprint profile.

Scenario	Condition	Mutual visibility % [Δ pp]	Coordination-nurse participation (%)	Δ 90% footprint pp [95% CI]
Normal	Baseline	1.04 [ref.]	38.6	n/a
Normal	COCPIT	2.55 [+1.51]	50.5	-1.5 [-2.0, -1.1]
Normal	NURSTA	0.84 [-0.20]	38.3	-0.5 [-1.0, 0.1]
Normal	Both	2.12 [+1.08]	50.7	-2.0 [-2.4, -1.5]
High	Baseline	0.72 [ref.]	44.3	n/a
High	COCPIT	1.80 [+1.08]	57.3	-1.6 [-2.0, -1.2]
High	NURSTA	0.58 [-0.14]	43.9	-0.3 [-0.7, 0.1]
High	Both	1.43 [+0.71]	56.6	-2.0 [-2.3, -1.6]

Note. Mutual visibility is reciprocal-perception time divided by every possible staff-pair second during the final ten-hour analysis window. Values in brackets are percentage-point changes from the same-scenario Baseline.

The corresponding COCPIT fold changes are 2.45 under normal load and 2.50 under high load. Folds are secondary because a baseline near 1% magnifies relative percentages. This is not a local co-presence measure. Coordination-nurse participation is the share of validation-counted staff contacts involving that role. Footprint change is relative to the same-scenario Baseline. Bold marks the scenario maximum for visibility or participation and the largest footprint contraction.

Table 16. Local sensitivity factors and levels.

Factor	Low	Default	High	Function
Visibility response	0.07	0.14	0.21	Uptake
Crowding threshold	0.50 / 0.40	0.62 / 0.50	0.74 / 0.62	Suppression
Repeat-contact delay	90 s	180 s	360 s	Recontact
Station-return tendency	0.20	0.42	0.70	Return

Note. Crowding-threshold values are reported as normal load / high load. Bold values are the manually developed baseline settings.

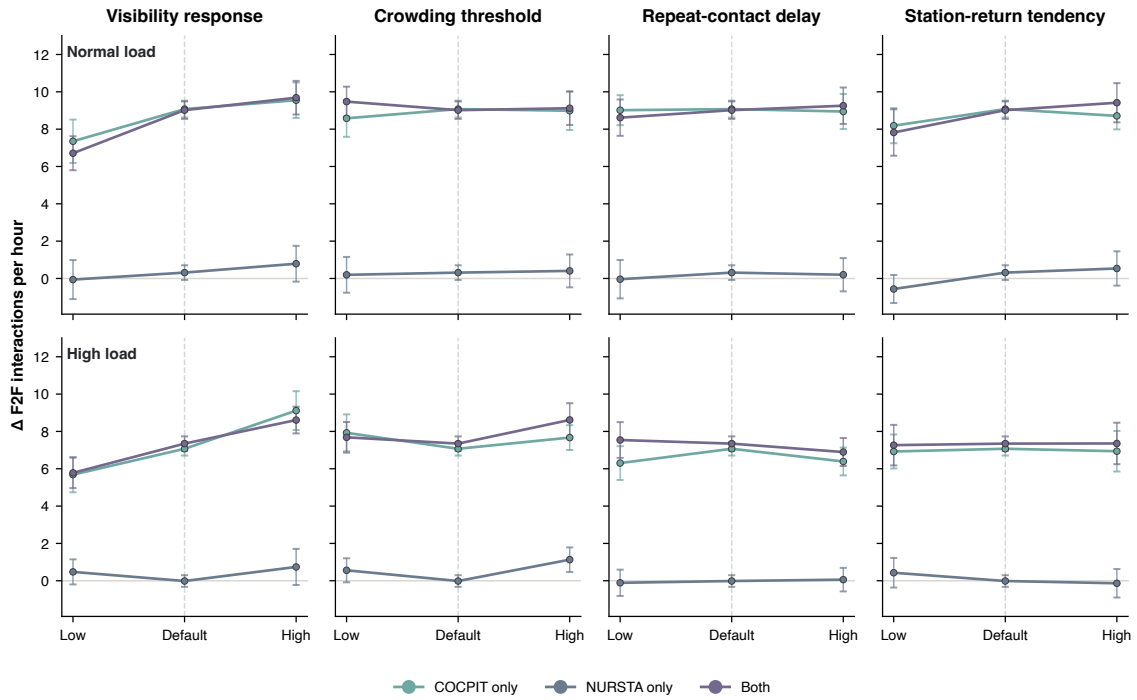


Figure 15. Local sensitivity response across four active behavioural parameters and two workloads.

Table 17. Local co-presence, co-awareness, and face-to-face contact under three visibility settings.

Setting	Nearby pair time (%)	Nearby pairs visible (%)	Colleague in view (%)	Δ F2F/h [95% CI]
Restricted	26.44 → 27.12	2.13 → 5.33	4.48 → 13.74	+6.57 [5.19, 7.96]
Default	45.62 → 46.35	2.03 → 5.05	7.16 → 15.82	+9.19 [8.50, 9.88]
Expanded	59.41 → 59.36	3.44 → 10.55	14.78 → 35.31	+11.90 [10.64, 13.17]

Note. Arrows show Baseline to COCPIT only means under normal load. Each setting used 20 paired seeds. Restricted, default, and expanded settings jointly used maximum sight distances of 12, 18, and 24 m, fields of view of 180, 210, and 270 degrees, and approach radii of 8, 10, and 12 m. Nearby pair time is the share of all staff-pair samples within the corresponding approach radius. Mutual visibility among nearby pairs is the share of those samples in which both staff members could see one another. Colleague in view is the share of staff time with at least one visible colleague. The final column is the paired COCPIT-minus-Baseline effect.

E Persona mechanism and experience check

The engagement-rate denominator changed across conditions because spatial changes generated different numbers and kinds of sampled opportunities. Counts and the main opportunity categories are therefore reported in Tables 18 and 19 before the experience check.

Table 18. Engaged decisions and sampled opportunities by persona.

Scenario	Persona	Baseline engaged / sampled	Both engaged / sampled
Normal	Adaptive Generalist	273 / 482	423 / 918
Normal	Focus Protector	24 / 452	27 / 1,039
Normal	Patient Advocate	100 / 500	166 / 1,023
Normal	Team Connector	216 / 483	379 / 897
Normal	Vigilant Monitor	189 / 474	408 / 903
High	Adaptive Generalist	121 / 178	207 / 515
High	Focus Protector	15 / 266	23 / 633
High	Patient Advocate	63 / 194	124 / 579
High	Team Connector	134 / 195	228 / 591
High	Vigilant Monitor	143 / 206	342 / 530

Table 19. Sampled opportunity mix at Baseline and under the combined spatial intervention.

Scenario	Condition	Bed flow	Handoff	Post-task	Station check-in	Sampled n
Normal	Baseline	35.1%	26.3%	3.8%	34.4%	2,391
Normal	Both	45.6%	29.5%	7.3%	17.5%	4,780
High	Baseline	58.8%	32.3%	6.6%	2.0%	1,039
High	Both	58.7%	29.2%	11.2%	0.9%	2,848

Note. Percentages are pooled across five designed personas and ten seeds. Rare corridor-passing and high-acuity categories are omitted from the displayed percentages but retained in the total (n).

Table 20. Engagement rates after standardising coarse opportunity composition.

Persona	Load	Baseline (%)	Both (%)	Δ pp	95% CI	Common support (%)
Team Connector	Normal	51.1	38.3	-12.8	[-20.1, -6.0]	99.8 / 99.8
Focus Protector	Normal	5.4	2.5	-2.9	[-5.9, -0.6]	100.0 / 99.1
Patient Advocate	Normal	22.2	14.4	-7.8	[-11.6, -4.9]	100.0 / 99.8
Vigilant Monitor	Normal	45.0	40.9	-4.1	[-8.6, 0.2]	99.8 / 99.1
Adaptive Generalist	Normal	57.9	46.6	-11.4	[-16.3, -6.4]	99.2 / 100.0
Team Connector	High	68.0	39.8	-28.2	[-37.5, -19.5]	99.5 / 100.0
Focus Protector	High	6.3	3.4	-2.9	[-7.6, 0.9]	99.2 / 98.9
Patient Advocate	High	34.5	19.3	-15.1	[-23.1, -7.5]	100.0 / 99.3
Vigilant Monitor	High	70.5	63.5	-7.0	[-12.1, -1.4]	99.5 / 98.5
Adaptive Generalist	High	66.2	41.3	-24.9	[-30.9, -17.2]	99.4 / 98.8

Note. Values are percentages of sampled opportunities; Δ is Both minus Baseline in percentage points, with 95% paired seed-bootstrap intervals. Within each workload and persona, both conditions were standardised to the same distribution of receiving staff role, model work reason, and partner role. Opportunity types absent from either condition were excluded. Common support retained 98.5–100% of sampled decisions. Common-support percentages are shown as Baseline / Both. This secondary diagnostic controls coarse composition, not complete decision situations.

Every staff identity received each persona once per scenario-condition-seed block. OCEAN personality-trait displays were derived only for public explanation and were excluded from prompts and inference. Language-model opportunities were sampled with a condition-blind deterministic hash and constrained by per-run, per-agent, and temporal-window budgets. The fixed model revision was 995ad96eacd98c81ed38be0c5b274b04

031597b0. Inference used vLLM, schema-constrained JSON, non-thinking mode, greedy decoding, and temperature 0.0.

Memory retrieval admitted at most five realised source-linked interactions from the previous two hours. Candidates required an explicit colleague, patient context, reason, interaction context, or recent-zone match. Four bounded values—coordination need, interruption strain, task continuity, and team support—were updated every two hours from recent persona decisions and realised contacts. The next update and later choices received those values together with their supporting event identifiers. Generated prose, missed opportunities, and post-run questionnaire answers were excluded.

Table 21. Choices replayed without accumulated experience.

Persona	Decisions replayed	Changed without accumulated experience
Team Connector	100	16 (16%)
Focus Protector	100	8 (8%)
Patient Advocate	100	13 (13%)
Vigilant Monitor	100	7 (7%)
Adaptive Generalist	100	9 (9%)
All personas	500	53 (10.6%)

Note. Each completed-run choice was replayed with the same persona, opportunity, and recent realized contacts but without the four accumulated-experience values. This checks that accumulated experience entered later decisions. It does not estimate an effect on real staff behaviour.

E.1 Common-situation checks of the five designed persona policies

The same 120 accepted decision situations were replayed for all five personas. Their actions differed in 96 situations (80%), and all five prespecified comparisons followed the expected direction.

Table 22. Common-situation checks of the five designed persona policies.

Prespecified comparison	Shared situations	Estimated contrast	95% CI
Coordination: Team Connector vs Focus Protector	101	+36.6 pp	[+27.7, +46.5]
Patient-linked: Patient Advocate vs others	120	+40.6 pp	[+29.1, +52.1]
Urgency: Vigilant Monitor vs others	120	+47.5 pp	[+36.6, +58.9]
Focus Protector: high vs low diversion	120	-25.0 pp	[-38.1, -12.5]
Adaptive Generalist: high vs low diversion	120	-50.2 pp	[-68.6, -30.3]

Note. The same accepted decision situation was replayed for every persona. Contrasts are paired engagement-rate differences, except the patient-linked and urgency rows, which compare each focal persona's contextual difference with the mean difference of the other four personas. Intervals use paired bootstrap resampling of situations. All five comparisons followed their prespecified direction.

F Synthetic appraisal audit

The appraisal sample comprised 100 paired staff units followed across four conditions. The final archive contained 800 of 800 expected responses with no schema or automated grounding errors. Condition names were omitted from the public-facing answers, but the supplied shift evidence could still reveal the layout indirectly. Neither a rule-blind re-rating of the

same traces nor a condition-label swap was run. Semantic correction was required for 534 packets (677 retries), but no packet exhausted the retry allowance.

The prespecified qualitative sample contained 40 bundles and 240 answers. A separate Codex audit retained 140 answers, excluded 100, and marked 109 as possible display excerpts. Thirty-two bundles contained at least three retained answers, and one persona-condition cell had no retained display response. No independent human coding or inter-rater reliability assessment occurred. Privacy/control was rateable in only 38 of 400 survey packets, and only three paired contrast cells were complete, so it was not treated as a main outcome.

Table 23. Recurrent design suggestions in retained synthetic counterfactual answers.

Synthetic design suggestion	Retained answers, n (%)	Personas represented
Visual permeability or awareness	14/37 (37.8%)	5/5
Semi-private coordination space	14/37 (37.8%)	4/5
Distributed support near active work	14/37 (37.8%)	4/5
Availability or focus signalling	10/37 (27.0%)	5/5
Acoustic control or quiet	3/37 (8.1%)	2/5

Note. Separate Codex coding retained 37 grounded counterfactual answers from the 40 prespecified bundles. Categories overlap. These are post-hoc model-assisted themes, not human thematic analysis or Zurich ED staff preference evidence.

The [interactive companion explorer](#) presents condition-specific quantitative profiles and only retained synthetic excerpts. Its characters and OCEAN displays are explanatory devices and do not enter the scientific analysis.

Table 24. Inclusive synthetic person-space fit contrasts.

Scenario	Contrast	Δ average fit [95% CI]	Δ fit floor [95% CI]	Δ persona SD [95% CI]
Normal load	COCBIT only – Baseline	+0.4 [0.2, 0.6]	+0.5 [0.1, 0.9]	-0.15 [-0.36, 0.07]
Normal load	NURSTA only – Baseline	+0.1 [-0.2, 0.3]	-0.2 [-0.8, 0.4]	0.00 [-0.33, 0.32]
Normal load	Both – Baseline	+0.3 [0.0, 0.6]	+0.3 [-0.6, 1.2]	-0.12 [-0.54, 0.30]
High load	COCBIT only – Baseline	+0.8 [0.4, 1.2]	+1.1 [0.2, 2.0]	-0.28 [-0.55, -0.02]
High load	NURSTA only – Baseline	+0.2 [-0.1, 0.5]	+0.5 [-0.3, 1.3]	-0.27 [-0.56, 0.02]
High load	Both – Baseline	+0.4 [0.1, 0.7]	+0.5 [-0.3, 1.3]	-0.10 [-0.46, 0.27]

Note. The fit floor is the lowest of the five persona scores within each seed. Persona SD is the within-seed standard deviation across the five personas. A negative change indicates less disparity. Confidence intervals use seed-paired means (n=10). These are synthetic persona metrics, not population accessibility estimates.

Bold values have 95% confidence intervals that exclude zero.

F.1 Representative synthetic responses

One grounded high-load response to the same design-change question is reported for every persona. These responses are synthetic outputs generated from completed simulation records.

Table 25. Representative grounded responses from the five model personas.

Persona	Representative synthetic response
Team Connector	“I would add fixed, semi-private standing areas within the shared circulation zones. This would allow for brief, focused handoffs without requiring a trip to the nursing work area or COCPIT station. It would reduce the missed opportunities caused by colleagues moving away, ensuring that shared awareness is maintained even in high-traffic areas.”
Focus Protector	“I would add a visual status indicator at the entrance to the patient rooms or in the circulation area to show when a nurse is in a focused task state versus available for coordination. This would help colleagues decide whether to approach, reducing the missed opportunities in the corridor and allowing me to protect my focus without losing critical connections.”
Patient Advocate	“I would introduce designated, semi-private huddle zones within the shared circulation area. These would provide a clear, accessible space for quick handoffs and patient status updates, reducing the need to travel back to COCPIT or wait for colleagues in open corridors. This would enhance continuity and explanation without sacrificing the openness of the circulation area.”
Vigilant Monitor	“I would add a digital status board at the NURSTA station that updates in real-time with nurse availability and patient acuity. This would allow me to scan for weak signals and coordination needs without having to physically approach the desk, which is currently an obstacle. This change would support my priority of risk-sensitive, evidence-focused coordination.”
Adaptive Generalist	“I would redesign the NURSTA nurse station to remove the relocated desk obstacle and integrate the standing area more seamlessly with the main flow. This would increase mutual visibility and reduce the missed-opportunity rate for spontaneous coordination. The trade-off is that it might increase interruptions for staff who are trying to focus on documentation or complex tasks, requiring stronger focus-protection signals.”

Note. These are synthetic responses generated from completed simulation records. They are not staff testimony.

G Comparison with the observed questionnaire

Table 26. Observed end-of-shift questionnaire and synthetic construct crosswalk. English wording is a concise translation of the German form.

Item	Observed end-of-shift construct	Scale	Synthetic evidence mapping
Q1	Maximum workload relative to other days	1–7	Peak demand, occupancy, waiting, task, and interruption traces
Q2	Shift share at critically high workload	0, 25, 50, 75, 100%	Critical-window share. The most demanding quarter anchored the aggregate
Q3	Ability to concentrate	1–7	Task continuity, interruptions, deferred contact, and unfinished work
Q4	Communication satisfaction across the ED team	1–7	Realized coordination, missed opportunities, updates, and handoffs
Q5	How well the team worked hand in hand	1–7	Cross-role coordination, completed handoffs, and workflow continuity
Q6	How well the ED helped patients	1–7	Operational proxy from patient-facing work and service progression; no care outcome
Q7	Information availability for working well	1–7	Information-bearing updates, handoffs, retrieval, and unresolved needs
Q8	Perceived noise	1–7	Unsupported: no acoustic state. Refusal required

The separate baseline questionnaire comparison comprised 300 normal-load shift packets and 90 evidence-control packets. All 390 passed schema and completeness checks. Q1–Q7 were rateable from supported traces. Q8 and all evidence-withheld controls correctly refused unsupported ratings. High load increased maximum workload and critical-workload share, satisfying the directional stress check.

Observed values were withheld from inference and compared locally afterward. Across the six scored 1–7 items, mean absolute error was 0.23 points and five means met the pre-specified half-point margin. Distributional compatibility passed for four of seven items. Relationship profiles correlated at 0.45, with a 0.30 correlation-matrix mean absolute difference and agreement on four of five core signs, but neither the distributional nor relationship gate passed and the complete-case sample was only 13 shifts. The result is therefore limited to descriptive similarity in group means rather than held-out validation. Q2 was excluded from the mean comparison because its construction partly fixed the ordinary-load mean near 25%. Shift-level variation and the high-load increase remained informative.